A Project Plan

On

DEEP LEARNING FOR AI-GENERATED IMAGE DETECTION: A REVIEW OF CNN, VISION TRANSFORMER, AND EXPLAINABLE APPROACHES

Submitted to

Amity University Uttar Pradesh

in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

Computer Science and Engineering

by

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October 2026

DECLARATION

We, Rohit Yadav (A2305223472), Vishal (A2305223568) and Hardik Mehlawat (A2305223449), students of B.Tech (Computer Science and Engineering) hereby declare that the project titled "Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches" which is submitted by us to Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

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CERTIFICATE

On the basis of declaration submitted by Rohit Yadav, Vishal and Hardik Mehlawat, students of B. Tech (Computer Science and Engineering), I hereby certify that the project titled "Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches" which is submitted to Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is an original contribution with existing knowledge and faithful record of work carried out by them under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

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Date: _____

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ABSTRACT

This report presents the Semester 7 minor project, a systematic review of deep learning methods for detecting AI-generated still images. Cheap, publicly available text-to-image generators now produce pictures that are ever harder to tell from photographs by eye. The resulting misinformation, fabricated evidence, impersonation and contamination of web-scraped image collections outpace manual inspection, making automated detection a practical necessity. Under a search protocol fixed before collection, eight query seeds were run through the search sources the protocol names. The 48 primary studies, published from 2019 to 2025, were each read in full, with their details recorded in a fixed-field extraction table; video and audio deepfakes were excluded. Grouped by where a detector's evidence comes from, the surveyed approaches form five families: handcrafted and frequency-domain forensics, convolutional neural networks adapted by transfer learning, transformer and other attention-based models, detectors that reuse frozen vision-language features, and explainability-focused work. Benchmarks, datasets and evaluation metrics receive the same comparative treatment. The main finding is that cross-generator generalisation is a structural problem, since a large part of what a detector learns reflects a single generator's design. Reported accuracy drops from 99.9% on a detector's training generator to 54.9% on one unseen generator, and from 92.77% to 65.77% when realistic in-the-wild images replace a standard benchmark. Three properties recur in the strongest cross-generator results: frozen general-purpose representations, inputs derived from the image, and diverse training data. Of the 48 studies, 36 report a cross-generator evaluation and four an explanation analysis, yet almost none combine the two. This gap motivates the experimental study proposed for Semester 8.

Keywords: AI-generated image detection; transfer learning; convolutional neural networks; vision transformer; explainable AI; synthetic image forensics

1. INTRODUCTION

1.1 Purpose of Plan

This is the final report of the minor project carried out by Group 148 in the seventh semester of the B.Tech programme in Computer Science and Engineering. It was undertaken as the Non-Teaching Credit Course (NTCC) Minor Project, ETMN100, and ran for twelve weeks, from 20 July to 9 October 2026, guided by Dr. Richa Gupta. The report follows Amity's project plan format, so it describes the goals, scope, requirements and management of the project as well as the work itself. That work was a systematic review of deep-learning methods for detecting AI-generated still images, that is, for deciding whether a single image was taken by a camera or produced by a generative model.

The report explains why the review was needed and how it was planned and carried out: a search protocol fixed in the first week, screening of the collected records, full-text extraction of the 48 primary studies into a structured table, and a check of every reference identifier against arXiv, Crossref or proceedings records. It then reports what the comparison of published results showed, especially for generators missing from a detector's training data, and uses the gaps it exposed to shape the experimental study proposed for the Semester 8 major project.

Every experimental result in this report was published by other researchers. No experiment was run for this review, and no model was trained or tested for it; each performance figure quoted is the value reported by its cited authors.

1.2 Background Information and Available Alternatives

The first models in wide use for photograph-like images were generative adversarial networks (GANs), introduced in 2014, which train a generator network against a discriminator network (Goodfellow et al., 2014). Later refinements, StyleGAN2 among them, improved GAN output until synthetic human faces could survive close scrutiny (Karras et al., 2020). Diffusion models brought the next shift: instead of pitting two networks against each other, they learn to undo the gradual noising of an image (Ho et al., 2020). Latent diffusion moved this process into a compressed latent space, making text-guided generation cheap enough for ordinary computers and for open release (Rombach et al., 2022). A convincing fake photograph

therefore no longer needs specialist ability; all it takes is a few seconds and a written prompt. Such pictures already spread across public social media in numbers large enough to be compiled into research datasets (Chen and Zou, 2023).

These capabilities cause real, not hypothetical, problems. A generated image can lend weight to a false claim in the news or serve as fake evidence of an event that never happened. It can also be used to impersonate people, since generated faces are now convincing enough to have their own detection literature (Wang et al., 2023b). These threats fall within media forensics, an older field that until recently dealt mainly with edited images rather than fully synthesised ones (Verdoliva, 2020). Less visibly, large web-scraped image collections can no longer be taken to hold camera output alone, because synthetic images are posted on the very platforms those collections are drawn from (Chen and Zou, 2023). For these reasons, recent surveys treat AI-generated media detection as a research problem in its own right (Lin et al., 2024).

Checking images by eye cannot keep pace. A large platform receives more uploads in a single day than human reviewers could ever examine. In addition, the cues that detectors exploit are often invisible to a viewer. Many of these cues are statistical traces of the upsampling steps inside the generator, and they show up in the frequency spectrum but not in the picture itself (Zhang et al., 2019; Durall et al., 2020). For both reasons, the only practical option is automated detection.

A practitioner can choose among five families of automated detection. Handcrafted forensics uses hand-designed statistical or frequency-domain features. CNN-based methods train convolutional neural networks (CNNs) for the task, usually by adapting a network pretrained on a large general image collection, which is called transfer learning. Vision transformers rely on attention instead of convolution. Multimodal detectors reuse features of foundation models such as the vision-language model CLIP. Explainability-oriented work, using tools such as Grad-CAM, shows which image regions drive a decision. Chapter 9 examines each family.

For the project itself, the order of work was set by the Semester 7 scope rule, which makes the minor project a literature review; the experiment planned for the Semester 8 major project is built around the gaps that this review documented.

1.3 Statement of the Problem

Automated detection remains an unsolved research problem. A detector with a mean accuracy of 92.77% on a standard benchmark reaches only 65.77% on a benchmark assembled deliberately from realistic in-the-wild images (Yan et al., 2025a).

The main obstacle is generalisation from one generator to another. A detector tends to perform well on the generator family it was trained on and much worse on other families. Early results looked promising: Wang et al. (2020) reported 90.8 mAP (mean average precision), averaged over eleven test sets, ten of them from generators unseen during training, for a ResNet-50 trained on ProGAN images alone, while cautioning that the result held "for now". Diffusion models largely put an end to this transfer. Detectors trained on GAN images lose, on average, 15.2% AUROC (area under the receiver operating characteristic, or ROC, curve) on diffusion images (Ricker et al., 2024b). In another evaluation, four GAN-trained detectors tested on twelve generators, five of them diffusion models, averaged roughly 52–76% accuracy on uncompressed images and about 50–61% after the resizing and compression that a social platform applies, with the diffusion models among the hardest cases (Corvi et al., 2023b). The pattern recurs among diffusion models themselves, where a ResNet-50 that is 99.9% accurate on the generator it learned from reaches only 54.9% on another (Zhu et al., 2023). Part of the reason is that many cues used by detectors are tied to one generator's design, not to synthetic images as such. For example, a commonly used spectral cue can be eliminated by modifying the final upsampling layer of the generator, which drops detection from 84.8–99.8% to 0–0.3% (Chandrasegaran et al., 2021).

These results define the problem this review addresses. Published detectors are evaluated under different protocols, and they are rarely tested on unseen generators in the same study that analyses what drives their decisions. A consolidated and checkable picture of which detection approaches generalise is therefore missing. This review sets out to build that picture from the published evidence.

1.4 Organisation of the Report

Chapters 2 to 6 cover the planning of the project: goals and deliverables, scope and budget, constraints, requirements, and the timeline, roles and risks. Chapter 7 covers image generation by GANs and diffusion models, the origin of detectable artefacts and the position of this review among earlier surveys. Chapter 8 describes the

review method, from the search protocol to verification and quality control. Chapters 9, 10 and 11 present the taxonomy of detection approaches, the datasets and benchmarks, and the evaluation metrics. Chapter 12 compares the 48 primary studies. Chapter 13 discusses the open challenges and states the research gap, and Chapter 14 gives the conclusion, the limitations of the review and its future scope. Chapter 15 describes the individual contribution of the members. The references and a list of abbreviations (Appendix A) close the report.

2. GOALS AND OBJECTIVES

2.1 Project Goals and Objectives

The project aimed to collect, check and bring together the published evidence on deep-learning detection of AI-generated images, so that both what is known and what is still missing become clear. Two questions were given particular weight. The first is cross-generator generalisation: does a detector that learnt from the images of one generator still work on generators it never saw during training? The second is explainability: can a person see why a detector labelled an image as real or generated? The gaps that the evidence leaves open were then to shape the experiment that the team will run in its Semester 8 major project. The four broad objectives of the synopsis were broken down into nine concrete steps:

1. To write the search protocol, with its databases, query seeds and inclusion and exclusion criteria, before any literature was collected, so that later decisions followed fixed rules (Sections 8.1 and 8.2).
2. To run eight fixed query seeds, among them "synthetic image detection", "CIFAKE" and "cross-generator generalization", through the search sources named in the protocol (Section 8.1).
3. To screen every unique record on its title and abstract and to note how many records remained after each stage, from identification to final inclusion (Section 8.3).
4. To read each of the 48 primary studies in full and log it in an extraction table of eight fixed fields, from the detection approach and architecture to the reported accuracy or AUC (area under the ROC curve) and the key limitation (Section 8.4).
5. To sort the primary detection studies into a taxonomy of five families, running from handcrafted and frequency-domain forensics to work on explainability (Section 8.5 and Chapter 9).
6. To compare the datasets and benchmarks used to test detectors, and the metrics through which the surveyed studies report their results (Chapters 10 and 11).

7. To build a master comparison table whose *Cross-Generator Tested?* and *Key Limitation* columns show, study by study, whether a detector was tried on unseen generators and where it falls short (Chapter 12).
8. To analyse the open challenges and state the research gap clearly enough to define the Semester 8 study (Chapter 13 and Section 14.3).
9. To check the identifier of every included reference, whether a DOI, an arXiv ID or a proceedings page, so that the reference list holds no unverified entry (Section 8.6).

2.2 Project Deliverables

The project produced the deliverables below; for tables and figures that appear in this report, the chapter is given in brackets.

- This final report, prepared in the Amity report format in October 2026, once the twelve-week window had ended on 9 October.
- The master comparison table of the 48 primary studies (Chapter 12); a script compared each of its rows with the extraction records, and all 48 matched.
- The dataset comparison table, which sets 11 datasets side by side (Chapter 10).
- Three figures drawn as SVG vector graphics: the flow of records through search and screening (Chapter 8), the taxonomy of detection approaches (Chapter 9) and an overview of the evaluation metrics (Chapter 11).
- The screening and extraction records, kept as CSV files: the 75 screened candidates with the decision taken on each, and the eight-field extraction table for the 48 primary studies.
- A BibTeX database holding the 74 included references under their screening IDs; every one of its 74 identifiers was confirmed against an arXiv, Crossref or PMLR record.
- The twelve weekly progress reports on the official one-page proforma, the reports for the first and second progress reviews (Project Progress Reports 1 and 2, dated 27 August and 1 October 2026), and the NTCC project diary.

3. SCOPE

3.1 Scope Definition

The project covers a single problem: deciding whether a still image was produced by a generative model. Detection is treated as binary classification, in which the detector looks at one image and assigns it to one of two classes, real (captured by a camera) or AI-generated. No particular generator is singled out. The review takes in whichever generators the surveyed studies train and test on, which means generative adversarial networks (GANs), diffusion models and any further generators that appear in those studies' experiments; GANs and diffusion models are described in Chapter 7. On the detection side, the approaches in scope run from handcrafted and frequency-domain forensics, through CNN transfer learning and Vision Transformers, to detectors built on CLIP features and explanation methods such as Grad-CAM. The scope also takes in the datasets and benchmarks on which these detectors are evaluated and the metrics in which their results are reported.

A detection or benchmark study entered the review only if it met the inclusion criteria of the protocol. It had to deal with still images, to have been published between 2019 and 2026, to report quantitative results such as accuracy, AUC or F1, and to be either peer-reviewed or a well-cited preprint. The publication window binds these studies alone. Background references on generators, backbone networks and explanation methods are not held to it, which is why some of them date from 2014 to 2018.

Within these limits, the work of the project consisted of the following activities:

1. Defining the search protocol and its criteria.
2. Running the eight query seeds through the search sources named in the protocol.
3. Screening the records by title and abstract.
4. Extracting data from the full texts of the primary studies.
5. Building the taxonomy of detection approaches.
6. Comparing the datasets and benchmarks.
7. Reviewing the evaluation metrics and how studies report them.
8. Comparing the reported results across the primary studies.

9. Analysing the challenges and identifying the research gap.
10. Verifying the identifier of every included reference.
11. Writing the review and this report.

3.2 Limitations of Scope

Some boundaries were drawn on purpose. Deepfake detection in video and in audio is left out, and studies that deal only with those media were filtered out when the search was run. The project neither builds nor deploys a detection system, so it produces no app and no real-time tool. The review itself also contains no experiment. Every accuracy, AUC or other performance figure quoted in this report is the value published by the original authors, and none of these values has been reproduced by the team. Experiments on the problem are reserved for the Semester 8 major project.

Other boundaries come from the protocol and from the timing of the search. Papers not written in English, papers whose full text could not be accessed and studies without a quantitative evaluation were all excluded. The well-cited rule for preprints also led to the exclusion of one candidate, a preprint with only a single citation. Because the search was run in Week 2 (27 July to 2 August 2026), research that appeared after that week is not part of the surveyed set.

Finally, the included studies did not all receive the same depth of treatment. The 48 primary studies were each read in full and extracted field by field. The 11 secondary studies were drawn on selectively and did not go through full-text extraction, so none of them has a row in the master comparison table of Chapter 12. Section 14.2 discusses the limitations of the review as a whole.

3.3 Projected Budget

Because the project reads and analyses published work instead of building a system, nothing had to be bought to carry it out: no hardware, no software licence, no dataset and no database subscription. The resources it did rely on are described here along with the extra money each one required.

Papers were read in freely accessible versions wherever such versions existed, for example arXiv preprints and open-access proceedings, and the team paid for no subscription. The software was either free or already installed on the machine in use (Section 5.3). The review work needed no GPU, so the team's existing laptop was

enough (Section 5.4). Internet access and electricity came from the members' existing personal resources. The one expense still ahead is printing and binding: the Amity guidelines ask for two spiral-bound copies of the report with a blue back cover, and that cost falls at submission. Since the amount is not yet known, Table 3.1 leaves it open instead of guessing a figure.

Table 3.1. Estimated additional cost of the project

Resource	**Use in the project**	**Additional cost (Rs.)**
Literature	Free versions of papers where available; no paid subscription	0
Software	Free tools or software already installed	0
Hardware	Existing laptop; no GPU needed	0
Internet and electricity	Existing personal resources	0
Printing and binding	Two spiral-bound copies with blue back cover, paid at submission	Not yet known
Total additional development cost	Excluding printing and binding	0

4. CONSTRAINTS

4.1 Project Constraints

Nine constraints shaped how the project was planned and carried out; they arise from the NTCC course guidelines, the review protocol, the Semester 7 scope rule that makes the deliverable a literature review, and the state of the published literature.

1. **Fixed twelve-week window.** The NTCC Minor Project (ETMN100) ran in a fixed window of 82 days, from 20 July to 9 October 2026, divided into twelve weeks running Monday to Sunday, with a short final week from 5 to 9 October. For every week a one-page Weekly Progress Report (WPR) naming all three members was due on Amizone by the following Wednesday, and the eligibility rule asks for 90% of the WPRs to be submitted and 80% of them to be satisfactory. The work therefore had to be broken into weekly steps that each ended in something reportable; Section 6.1 gives the resulting timeline.
2. **No experiments in the minor project.** The Semester 7 scope rule limits the minor project to a literature review, so it runs no experiments and reports no results of the team's own. Every accuracy, AUC or other score quoted in this report is the figure its original authors published, obtained with their own data, settings and test protocols; the review can check that a figure has been copied faithfully from the paper, but it cannot reproduce it. Running the experiments that the review argues for is the task of the Semester 8 major project (Section 14.3).
3. **Inconsistent reporting.** The surveyed studies report different metrics, among them accuracy, average precision, mean average precision and AUC, computed over different sets of test generators, so two headline figures from different papers seldom measure the same thing. Since one metric cannot be rebuilt from another once a paper is published, and threshold-based figures such as accuracy usually rest on a default operating point, many results can be placed side by side but not ranked against each other (Section 11.2).
4. **Access to full texts.** The protocol excluded any study whose full text could not be accessed, and every primary study was analysed from its full text rather than its abstract. Relevant work that the team could not open in full therefore had to

stay outside the review. The protocol also limited the review to papers written in English.

5. **A fast-moving field.** Because the field moves quickly, the generator lists and benchmarks used in detection studies go out of date soon after they appear (Section 13.3). The database searches were run in the literature collection round of Week 2 (27 July to 2 August 2026), and work that appeared after that round is not covered. Preprints were admitted only if they were well cited, and one preprint with a single citation was excluded for failing this test.
6. **Citation integrity.** No reference could be cited until its identifier, whether a DOI, an arXiv record or a proceedings page, had been opened and its title and first author matched against the screening record. All 74 references in the project bibliography passed this check, through the arXiv metadata record, the Crossref record of the DOI or, in one case, the PMLR page of the proceedings, and the same pass found and corrected one wrong publication year (Section 8.6).
7. **Originality.** The Amity guidelines require the report to be written by the students in their own words and allow a similarity index of no more than 15% on Turnitin or a similar tool; anything over that limit has to be rewritten. The guide first screens the report with open-source software and the final version then goes through Turnitin on Amizone, so the report has to be the team's own writing, with every source it draws on cited.
8. **Report format.** Presentation follows the Amity report format: English text on A4 paper, side margins of 1.25 inch and top and bottom margins of 1.0 inch, line spacing of 1.5, and references cited in the text by surname and year and listed alphabetically. Within those margins the two widest tables, the dataset comparison in Chapter 10 and the master comparison of the 48 primary studies in Chapter 12, would not fit across a portrait page, so both are set in landscape orientation.
9. **Computing resources.** An ordinary laptop with an internet connection was enough for searching, reading, drafting and checking (Section 5.4), because a literature review trains no models and so needs no GPU. No hardware was bought for the project.

5. REQUIREMENT ANALYSIS

In a software project, requirement analysis sets out what a system must do and the conditions it has to meet. The minor project builds no system; what it produces is a systematic review of published work on AI-generated image detection, together with the report that presents it. The requirements in this chapter therefore describe what the review process and its outputs must deliver. The functional requirements in Section 5.1 name the tasks the review has to carry out, the non-functional requirements in Section 5.2 set the qualities its outputs must have, and Sections 5.3 and 5.4 list the software and hardware the work needed.

5.1 Functional Requirements

- **FR1 – Search.** The review must run the eight query seeds recorded in the search protocol through the search sources the protocol names, and keep each candidate together with the page where its existence was confirmed (Section 8.1).
- **FR2 – Screening.** Every record found must be de-duplicated and then screened by its title and abstract against the inclusion and exclusion criteria fixed before collection began, with the number of records logged at each stage (Sections 8.2 and 8.3).
- **FR3 – Extraction.** For each primary study the eight extraction fields must be filled from the full text, and any cell that cannot be confirmed must be marked for verification rather than estimated (Section 8.4).
- **FR4 – Taxonomy.** Each primary detection study must be assigned to one of the five families of the taxonomy in Chapter 9 (or to two, where it belongs to both), while studies centred on a dataset or benchmark are analysed in Chapters 10, 12 and 13 instead.
- **FR5 – Datasets.** The datasets and benchmarks used across the surveyed studies must be compared side by side in one table, and their known problems must be discussed (Chapter 10).
- **FR6 – Metrics.** Each evaluation metric the review relies on must be defined by its formula, with the AI-generated image taken as the positive class, so that every quoted figure has a stated meaning (Chapter 11).

- **FR7 – Master table.** All 48 primary studies must appear in one comparison table that includes the Cross-Generator Tested? and Key Limitation columns, with every cell drawn from the extraction records (Chapter 12).
- **FR8 – Gap analysis.** The research gap and the study planned for Semester 8 must be derived from the review's findings, with each element of the planned study justified by a finding and presented only as future work (Chapter 13 and Section 14.3).
- **FR9 – Bibliography.** The review must keep one bibliography of the included references, with each entry keyed by its screening ID, and no entry may be cited until its identifier has been opened and checked (Section 8.6).
- **FR10 – Reporting.** Progress must be reported every week through a WPR on the official proforma, and the final report must be delivered in the Amity format along with the Minor Project Diary and the full set of WPRs.

5.2 Non-Functional Requirements

- **NFR1 – Traceability.** Every number and claim taken from the literature must lead back to a cited source, and every cell of the master table must match the extraction records.
- **NFR2 – Verifiability.** Each reference must carry a DOI, an arXiv identifier or a stable proceedings page that an evaluator can open to confirm it.
- **NFR3 – Reproducibility of the search.** The databases, query seeds, criteria and counts at each stage must be recorded in enough detail for another reader to repeat the search and the screening.
- **NFR4 – Consistency.** The same eight extraction fields must be applied to every primary study, and each metric must keep a single definition throughout the report.
- **NFR5 – Originality.** The report must be written in the team's own words, with a similarity index of no more than 15% when checked through Turnitin.
- **NFR6 – Readability.** The text must be clear to an evaluator who is not a specialist in image forensics, with technical terms explained at first use and every table and figure numbered by chapter and listed in the front matter.

- **NFR7 – Timeliness.** Each WPR must reach Amizone by the Wednesday following its week, and the final report must be uploaded by the NTCC deadline of 26 October 2026.
- **NFR8 – Maintainability.** Working files must be kept in plain-text formats (Markdown drafts, CSV records and a BibTeX reference file) so that they can be edited, compared across versions and checked by script.

5.3 Software Requirements

Table 5.1 lists the software and tools the work relied on, from the literature search through to the final PDF; no software licence was bought for any of them (Section 3.3).

Table 5.1. Software and tools used in the project

Software / tool	Purpose
Scholarly databases and search engines named in the protocol (FR1)	Running the query seeds; confirming candidates on publisher and proceedings pages
CSV files	Candidate list, screening decisions and extraction table
BibTeX	Bibliography of the 74 included references, each keyed by its screening ID
arXiv, Crossref and PMLR records	Confirming each reference's identifier, title and first author
Visual Studio Code and Markdown	Editor and plain-text format for the chapter drafts
Python	Scripted checks, e.g. comparing the master table with the extraction records
SVG vector graphics	Drawing the three figures
Microsoft Word	Assembling the report and producing its PDF
Microsoft Edge (headless mode)	Turning each WPR proforma into a PDF
Poppler PDF tools	Checking PDFs for selectable text and embedded fonts
Git and GitHub	Version control in a private repository, adopted on 20 August 2026

5.4 Hardware Requirements

The review work was carried out on the laptop described in Table 5.2. Because a review trains no models, no GPU was needed, and the project bought no hardware (Section 3.3).

Table 5.2. Hardware used in the project

Component	**Specification**
Processor	Intel Core i5-1135G7
Memory	16 GB RAM
Operating system	Windows 10
Network	Ordinary internet connection
GPU	Not required for the review

6. PROJECT MANAGEMENT APPROACH

The project was run as twelve weekly stages that together covered the NTCC window, from 20 July to 9 October 2026. Each stage was planned around a defined piece of work. Its progress was then written up in a Weekly Progress Report (WPR) on the official one-page proforma, which names all three members, and each WPR was due on Amizone by the Wednesday following its week. The portal entries that went with the uploads also set out the target for the week, what had been achieved and the work planned next. The weekly reports also bear on eligibility, because the Amity guidelines require 90% of WPRs to be submitted and 80% to be judged satisfactory. Two progress reviews served as milestones beyond the weekly reports. The first was a review presentation on 27 August 2026, for which the NTCC project diary and Project Progress Report 1 were prepared. The second rested on Project Progress Report 2, which is dated 1 October 2026 and reports progress up to the end of Week 10. The work was carried out under the supervision of the faculty guide, Dr. Richa Gupta.

6.1 Project Timeline

Table 6.1 sets out the twelve weeks of the window with their dates, the activity planned for each and the deliverable recorded in its WPR, and closes with a row for the preparation of this report.

Table 6.1. Project timeline

Week	Dates	Planned activity	Deliverable
1	20–26 Jul 2026	Fix the title and scope; define the search protocol with its inclusion and exclusion criteria; set the section outline	Search protocol and outline
2	27 Jul – 2 Aug 2026	First round of literature collection; screening by title and abstract	Candidate and screening lists (101 records found, 75 unique)
3	3–9 Aug 2026	Full-text reading of the primary studies; data extraction	Extraction table covering all 48 primary studies
4	10–16 Aug 2026	Finalise the taxonomy; draft the introduction	Taxonomy assignment of the 48 studies; introduction draft
5	17–23 Aug 2026	Draft the background on GANs, diffusion models and their artefacts	Background draft; BibTeX file and verification log (74 of 74 verified)
6	24–30 Aug 2026	Draft the handcrafted and frequency-domain family and the CNN and transfer-learning family	First two taxonomy sections
7	31 Aug – 6 Sep 2026	Draft the transformer, multimodal and explainability families	Complete taxonomy draft

Week	Dates	Planned activity	Deliverable
8	7–13 Sep 2026	Draft the datasets and evaluation-metrics parts	Datasets and metrics drafts; dataset table (11 datasets); record-flow and metrics figures
9	14–20 Sep 2026	Build the master comparison table; draft the comparative analysis	Master table of the 48 studies, checked by script; comparative analysis draft
10	21–27 Sep 2026	Draft the challenges and gap analysis, the conclusion and the abstract	Challenges, conclusion and abstract drafts; review text complete, abstract included (about 9,850 words)
11	28 Sep – 4 Oct 2026	Assemble the full draft; verify the references; run consistency checks	Draft v1 (DOCX and 31-page PDF) citing 73 of the 74 included references
12	5–9 Oct 2026	Guide review of the draft; corrections; restyling in the Amity format; submission pack	No new deliverable; open items listed in WPR 12
After the window	October 2026	Final report prepared in the Amity format	This report

Every deliverable planned for Weeks 1 to 11 was completed and reported in the WPR for its week; as the dates in the NTCC project diary show, much of this work, the drafting in particular, was done ahead of this schedule. Week 12, the short final week, added no new deliverable. In it the submission pack was reviewed, the window closed on 9 October 2026, and four items were recorded as still open: the official report format, the result of the plagiarism check, the team's rewrite of the draft in its own words and a review of the draft by the guide. These open items were carried into the final submission, for which the NTCC calendar fixes 26 October 2026 as the date of upload on Amizone.

6.2 Project Roles and Responsibilities

The project was carried out by three students, Rohit Yadav, Vishal and Hardik Mehlawat. By the team's own decision, no work package was handed to a single member: every package was shared, and all three members worked on each of them. The packages follow the course of the review itself, from finding and screening the literature, through extracting and analysing it, to writing, checking and documenting the work. Table 6.2 lists them with the responsibilities that each one covered.

Table 6.2. Project roles and responsibilities

Role	Responsibilities	Participant(s)
Literature search and screening	Running the eight query seeds through the protocol's search sources; removing duplicates; screening titles and abstracts against the criteria	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)
Full-text reading and data extraction	Reading the 48 primary studies in full; recording eight fields for each; resolving doubtful cells	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)
Taxonomy, comparative analysis and gap analysis	Assigning the studies to five families; tallying the comparison table; stating the research gap	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)
Writing, tables and figures	Drafting the text; building the dataset and comparison tables; drawing the figures	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)
Reference verification and consistency checks	Opening and checking all 74 identifiers; script check of the master table against the extraction table	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)
Project documentation (WPRs, diary, progress reports, final report)	Weekly WPRs on Amizone; NTCC project diary; Project Progress Reports 1 and 2; this report	Rohit Yadav, Vishal and Hardik Mehlawat (jointly)

Decisions on the scope, on which studies to include, on the taxonomy and on the gap analysis were taken by the team together, under the direction of the guide.

6.3 Project Risk Assessment

A risk register lists the events that could stop a project from reaching its goals, grades each by how serious and how likely it is, and records the step taken to contain it; Table 6.3 gives the register for this project.

Table 6.3. Project risk assessment

Risk	Risk level (L/M/H)	Likelihood (L/M/H)	Mitigation strategy
Project size	M	M	Scope fixed to still images and to studies from 2019–2026; detailed comparison limited to 48 primary studies
Incomplete search coverage	H	M	Search sources and eight query seeds fixed in the protocol; candidates confirmed on publisher and proceedings pages; counts kept at each stage (101 records found, 75 left once 26 duplicates were removed)
Wrong or unverifiable citations	H	M	Only keys from the screening list cited; one BibTeX file keyed by screening ID; every one of the 74 identifiers opened and matched to its record (74 verified, none pending); one year error corrected

Risk	Risk level (L/M/H)	Likelihood (L/M/H)	Mitigation strategy
Transcription errors in reported figures	H	M	Figures copied only from the extraction table; a script compared the master table with it, and all 48 rows agreed; 15 dataset-table cells left blank where no value was recorded
Misreading what a study tested (for example, a cross-dataset test taken as cross-generator)	H	M	Ten doubtful extraction cells settled from the full texts; two cross-generator flags changed to No and those studies described accordingly
Inaccessible full texts	M	L	arXiv searched alongside publisher and proceedings pages, on which records were also confirmed; all 48 primary studies read in full
Inconsistent metrics across studies	M	H	Eight metrics defined with equations (Chapter 11); each study's reported figure kept as stated; mismatched metrics, thresholds and averages analysed (Section 11.2)
Fast-moving field	M	H	Studies up to 2026, arXiv included; papers found during writing screened before being cited; literature base unchanged from Week 9; staleness treated as a challenge (Section 13.3)
Timeline slippage	H	M	Twelve weekly stages with set outputs and per-section word budgets; a WPR by each following Wednesday; each drafting stage reviewed before its WPR; two progress reviews
High similarity index	H	M	Drafts treated as working versions pending a rewrite in the team's own words; an earlier draft sent for the departmental plagiarism check; final report to be checked on Turnitin against the 15% limit
Late availability of the report format	M	H	Draft deliberately kept in a clean generic style, ready for restyling; DOCX source retained; this report prepared in the Amity format after the window
Scope creep into experiments	H	L	Minor project limited to the review, with no experiments and no results of the team's own; planned experiments stated only as Semester 8 future work; drafts checked for own-results claims

Several of these mitigations were routine checks built into the work rather than one-off actions: each drafting stage from Week 4 to Week 11 was reviewed before its WPR was prepared, with citation keys compared against the screening list, numbers against the extraction table, and the text searched for any claim of results of the team's own. Some of the listed risks did occur. One publication year in the reference records

was wrong and was corrected; full-text checking changed the cross-generator flag of two studies to No, and the comparison uses these values; and the official report format was still unavailable when the window closed, so the final formatting was done in this report.

7. LITERATURE REVIEW

Before a detector can be understood, it has to be clear what evidence it looks for. This chapter supplies that background. It explains how generative adversarial networks (GANs) and diffusion models, the two generator families the literature concentrates on, produce an image, and why their output carries traces that a detector can use. It also positions this review against earlier surveys. The focus is on how generators work; the detection literature itself is analysed in Chapters 9 to 12.

7.1 Generative Adversarial Networks

In a GAN, two networks are trained in competition. The generator turns a vector of random numbers into an image, while the discriminator receives both real images and generated ones and has to tell them apart. Training alternates between the two, so the generator is never shown a particular image to reproduce; the only training signal available to it is the discriminator's current verdict on what it produced (Goodfellow et al., 2014). Nothing in this set-up requires the generated images to match real photographs statistically any further than is needed to stop the discriminator objecting, a point taken up in Section 7.3.

In structure, a GAN generator is a sequence of layers that upscales a small feature map stage by stage until full resolution is reached. Later architectures kept this basic form but altered its internals. StyleGAN2, for example, redesigned the normalisation and layer structure of the generator with the specific aim of removing characteristic artefacts seen in the output of the preceding version, and reported higher quality for high-resolution face synthesis (Karras et al., 2020). These changes had two effects. Synthetic faces became hard to recognise as fake by eye, and designing a generator turned partly into a matter of eliminating whichever traces had recently been pointed out.

7.2 Diffusion Models

A diffusion model does without the adversarial game and uses a two-part construction instead. In the forward process, a little Gaussian noise is added to a real image at each of many steps, until it can no longer be distinguished from noise. The forward direction is fixed, and nothing in it is learned. The model learns the inverse: shown a noisy image together with its step index, it is trained to estimate the noise that

had been added, which allows that step to be undone. To generate, the model starts with pure noise and keeps applying its learned reverse step until a picture emerges (Ho et al., 2020). As every step is only a minor correction rather than one large jump, training escapes the instability associated with adversarial optimisation.

Repeating the denoising step many times on full-resolution images demands heavy computation. Latent diffusion reduces this cost by running diffusion inside the latent space of an autoencoder trained in advance, where images are held in compressed form; diffusion works on a compact latent code, which a decoder finally maps back to pixels. The paper that introduced latent diffusion also brought in cross-attention conditioning, the means by which a text prompt can guide generation (Rombach et al., 2022). Many publicly released text-to-image systems rest on this design. For detection, an important feature is the division between a latent diffusion stage and the fixed decoder that follows it, as Section 7.3 explains.

7.3 Where Detectable Artefacts Come From

Both generator families build an image through repeated upsampling, and no other source of detectable traces has been studied as much. Convolutional generators that rely on up-convolution fail to match real images in their spectral distribution, and the mismatch is concentrated at high frequencies (Durall et al., 2020). Seen through the discrete cosine transform, the same defect takes the form of regular, grid-like patterns that recur in several GAN architectures (Frank et al., 2020). Direct evidence singles out the upsampling module as the cause: simulating that module alone, with no real generated images involved, can reproduce artefacts typical of a GAN pipeline (Zhang et al., 2019). These traces cannot be seen in the image itself; they belong to its frequency spectrum.

An important qualification applies: the traces stem from particular design choices, not from synthesis as such. If a generator's last upsampling layer is replaced by bilinear upsampling, the high-frequency spectral signature on which detectors had come to depend largely disappears (Chandrasegaran et al., 2021). Likewise, a spectrum-based classifier trained on the artefacts of one upsampler fails to generalise to GauGAN, which is built with a different upsampler (Zhang et al., 2019). Once such a trace is known, regularising the generator's spectrum during training can also deliberately suppress it (Durall et al., 2020). Frequency behaviour has accordingly been studied as

an attribute of generative models themselves, not merely as a handy cue for detection (Schwarz et al., 2021).

Looked at more closely, the trace identifies a single model rather than a family. Fingerprints learned from generated images are specific enough for an image to be attributed to the exact network that generated it, and models that differ only in their training run, for example in the random seed used at initialisation, are reported to leave fingerprints that can be told apart (Yu et al., 2019). There is also a hand-crafted equivalent: averaging the denoising residuals of many images from a single source, the same procedure that classical forensics uses to estimate a camera's sensor noise, gives a fingerprint capable of identifying the generating model (Marra et al., 2019). In both cases the measurement resembles a serial number more than a general trait of synthetic imagery.

Diffusion models alter this picture without wiping it out. Their output carries fewer detectable traces than GAN output and, under perturbation, lowers detection performance more readily (Ricker et al., 2024b). It is not trace-free, though. Forensic analysis of diffusion-generated images finds spectral fingerprints broadly of the same kind as those described above (Corvi et al., 2023b), and a comparison of fifteen models spanning GAN, VQ-based (vector-quantised), diffusion and latent-diffusion designs detects artefacts throughout: they differ between families, but none of the families is free of them (Corvi et al., 2023a). Latent diffusion contributes a further source, specific to its design. Because one fixed decoder is shared by all the images that a given system outputs, that decoder leaves a signature which training-free detectors are able to use (Ricker et al., 2024a); Section 9.2 covers such detectors. Lastly, every trace described so far is fragile. The traces sit at high frequencies, exactly the information that lossy compression discards; once JPEG compression is strong, a generated image's spectrum becomes essentially indistinguishable from that of a real one (Dzanic et al., 2020). Ordinary post-processing can also alter or conceal them (Corvi et al., 2023a).

This is why generalisation, introduced in Chapter 1, is a structural difficulty and not an engineering inconvenience. Suppose the evidence that a detector has learned stems from the upsampling operator, the training run or the decoder of one generator. A new generator differing in any one of these then does not just offer harder examples; it takes the evidence away. Much of what distinguishes the approaches reviewed in

Chapter 9 is how each tries to get around this dependence, and Chapter 13 comes back to it as the central open challenge facing the field.

7.4 Earlier Surveys and the Position of This Review

This review covers published research on the detection of AI-generated still images, within a deliberately narrow scope. It is restricted to single images, so deepfake detection in video and audio is excluded; to work published from 2019 to 2026; and to studies reporting quantitative results, so that the performance claimed in different studies can be set side by side. Chapter 8 gives the full search protocol and the inclusion and exclusion criteria.

Two features distinguish it from earlier reviews. First, previous overviews either take in more media, dealing with audio and video as well as images (Lin et al., 2024), or address a narrower subject, GAN-generated faces in particular (Wang et al., 2023b), or date entirely from the period before diffusion-based generation (Verdoliva, 2020). Second, this review records cross-generator evaluation and stated limitations for each study instead of leaving them as discussion points: for every study covered, the master comparison table in Chapter 12 notes whether its detector was evaluated on generators not seen in training and which limitation, if any, its authors acknowledge.

The review makes five contributions, each developed in a later chapter:

1. A taxonomy that groups detection approaches into five families: handcrafted, statistical and frequency-domain forensics; CNN-based approaches and transfer learning; transformer and attention-based approaches; multimodal and foundation-model-based approaches; and explainability in detection (Chapter 9).
2. A comparative overview of the datasets and benchmarks used in the field for training and testing, including ArtiFact, CIFAKE and GenImage (Rahman et al., 2023; Zhu et al., 2023; Bird and Lotfi, 2024) (Chapter 10).
3. An examination of reporting practice, showing that studies mix accuracy, AUC and average precision in varying combinations (Wang et al., 2020; Cozzolino et al., 2021; Zhu et al., 2023), a variation that limits how directly they can be compared (Chapter 11).

4. The master comparison table, whose dedicated *Cross-Generator Tested?* and *Key Limitation* columns let each headline figure be read together with the conditions under which it was obtained (Chapter 12).
5. A gap analysis (Section 13.6) motivating the second, planned phase of the project, outlined in Section 14.3: a transfer-learning comparison on CIFAKE (Bird and Lotfi, 2024) between four ImageNet-pretrained CNN backbones and a Vision Transformer benchmark (Dosovitskiy et al., 2021), with Grad-CAM (Selvaraju et al., 2017) to explain predictions and a subset of GenImage (Zhu et al., 2023) to test generalisation across generators.

This review itself conducts no experiments and reports none of its own; all performance figures in the chapters that follow are those reported by the cited authors.

8. REVIEW METHODOLOGY

This chapter sets out how the studies discussed in the rest of the report were found, filtered, read and checked. The procedure was written down as a protocol in Week 1 (20–26 July 2026), before a single paper had been collected, so that every record would be judged by the same rules and the process could be reported afterwards. The number of records entering and leaving each stage was logged as the work progressed, which allows the selection to be drawn as a PRISMA-style flow diagram.

8.1 Search Protocol

The protocol named six bibliographic databases: IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, arXiv and Google Scholar. The searches were driven by eight query seeds:

- "AI-generated image detection"
- "synthetic image detection"
- "GAN image detection"
- "diffusion image detection"
- "deepfake image forensics"
- "CIFAKE"
- "GenImage"
- "cross-generator generalization"

The search was organised into six topic strands that follow the structure of the taxonomy: GAN-era CNN detection; diffusion-era detection; frequency-domain and handcrafted forensics; ViT- and CLIP-based detection; datasets, benchmarks and surveys; and explainability together with background material. The searches themselves were carried out in Week 2 (27 July to 2 August 2026). The seeds were run against arXiv, against the pages of publishers and proceedings (IEEE, ACM, CVF, NeurIPS, ICML, ICLR, AAAI, PeerJ and Wiley) and against Google Scholar, and the raw list of hits for each strand was kept. Every candidate record stores the URL at which the paper's existence was confirmed. The year stored with a record is the year of its publication venue, which can be one year later than the paper's first appearance on arXiv.

8.2 Inclusion and Exclusion Criteria

Each record was judged against the four inclusion criteria and the four exclusion criteria laid down in the protocol. Table 8.1 places the two lists side by side.

Table 8.1. Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Published between 2019 and 2026	Deepfake work on video only or audio only
Deals with still images, not video or audio	No quantitative evaluation
Presents numerical results, for example accuracy, AUC or F1	Not written in English
Peer-reviewed, or a preprint that is well cited	Full text cannot be accessed

The date window applies to detection studies only. Background references on generator models, backbone networks and explanation methods are allowed to fall outside it by design, and some of these classic papers date from 2014 to 2018.

8.3 Screening and Selection

The searches returned 101 records. Once 26 duplicates had been removed, 75 unique records remained, and each was screened on its title and abstract against the criteria in Table 8.1. Apart from the single exclusion described below, out-of-scope items had already been filtered out at search time, so they were never counted among the identified records. In total, 74 records were included and divided into three groups:

- 48 primary studies on detection methods and datasets, each read in full (Section 8.4);
- 11 secondary studies, retained and drawn on selectively;
- 15 background references, covering generators, backbone networks, explanation methods and earlier surveys.

Only one record was excluded: a preprint with a single citation, which failed the well-cited criterion for preprints when citation counts were checked in Week 3. Figure 8.1 traces the records through each stage.

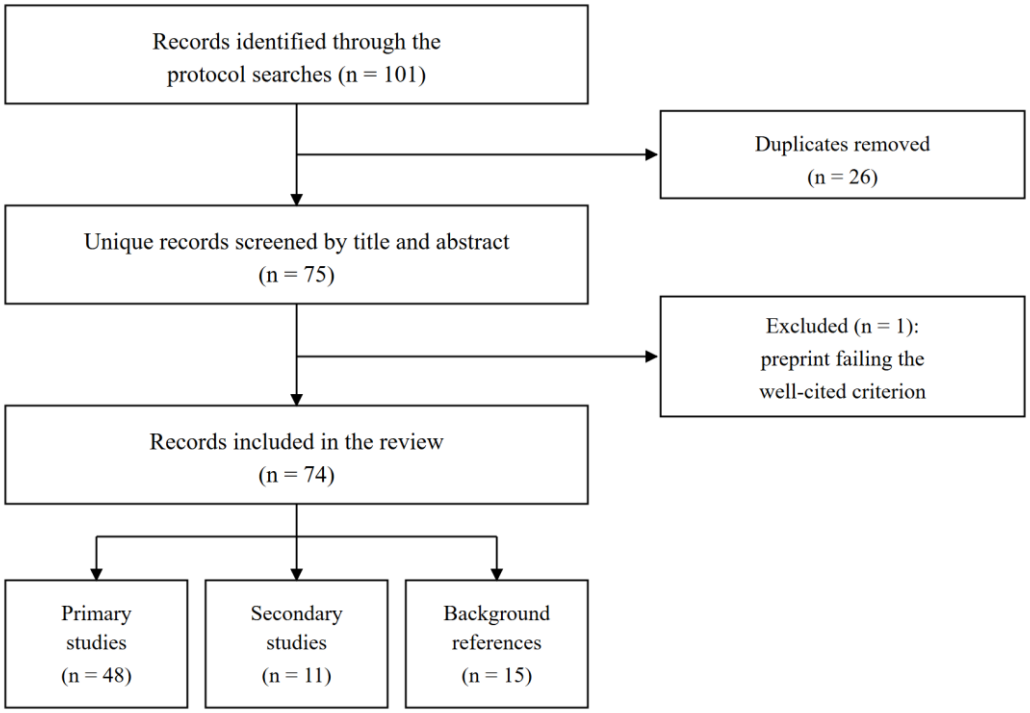


Figure 8.1. Records at each stage of the review, from the database searches to the final set: 101 identified, 26 removed as duplicates, 75 unique records screened on title and abstract, 1 excluded and 74 included (48 primary studies, 11 secondary studies and 15 background references).

8.4 Data Extraction

During Week 3 (3–9 August 2026), all 48 primary studies were read in full, and each was entered as one row of an extraction table. The table has eight fields, listed in Table 8.2. These fields had been fixed in the Week 1 protocol, before any paper was read, and they later became the columns of the master comparison table in Chapter 12.

Table 8.2. Fields recorded for each primary study

Field	What is recorded
Study (author, year)	First author and year of publication
Detection approach	A brief summary of the method or study design
Model or architecture	The network or classifier that makes the decision
Dataset used	The data used for training and testing
Generator(s) covered	The generators whose images the study uses
Reported accuracy / AUC	Headline result, in the metric the authors chose
Cross-generator tested?	Yes or No: is any test generator held out of training?
Key limitation	The main weakness that the authors themselves state; where they state none, the cell says so and may add the review's own note on the study's scope

When a value could not be confirmed from a paper's own text, its cell was flagged for verification instead of being filled with an estimate. Ten cells were flagged in this way. All ten were later settled from the full texts, and every settled value was then checked against its paper a second time, so the finished table contains no flagged cells.

Reading the full texts also changed the cross-generator flag of two studies to No. Frank et al. (2020) showed the frequency-domain artefact they studied across the GANs they analysed, but none of their experiments trains on one GAN and then tests on a GAN held out of training; the result on unseen architectures that they refer to comes from Wang et al. (2020). Aghasanli et al. (2023) ran a cross-dataset test that switched the image content between ImageNet and Oxford-IIIT Pet, yet the generated images on both sides came from one Stable Diffusion image-to-image process, so no unseen generator was involved. Neither study is counted as cross-generator evidence in Chapters 12 and 13.

8.5 Taxonomy Assignment

After extraction, the primary studies were placed in the taxonomy that Chapter 9 describes. Its five families are separated by where a detector's evidence comes from: handcrafted, statistical and frequency-domain forensics; CNN-based approaches and transfer learning; transformer and attention-based approaches; multimodal and foundation-model-based approaches; and explainability in detection. The assignment was recorded against each study in the screening records and was fixed in Week 4, before the text of Chapter 9 was drafted. Three studies sit across two families and were assigned to both. Four primary studies are concerned mainly with datasets and benchmarks rather than with a detection method; these were left outside the five families and are analysed in Chapters 10, 12 and 13 instead. The number of studies in each family is reported in Chapter 9.

8.6 Verification and Quality Control

Three kinds of checks were built into the process, so that each reference and each quoted value can be traced back to a source.

The first concerned citations. Before drafting began, a rule was fixed that drafts would refer to a study only through its screening identifier, and that a BibTeX file of

the included references would be keyed with the same identifiers. A study could be cited only if it had a row in the screening records and an entry in this file, and a paper discovered later, during writing, had to pass screening before it could be cited.

The second concerned the identifiers themselves. All 74 identifiers in the BibTeX file were opened and compared with an authoritative record: the arXiv metadata record for an arXiv identifier, the Crossref record for a DOI, and, for one paper without a DOI, the PMLR proceedings page. Titles and authors were compared with the screening rows, and for a DOI the venue, volume and pages were compared as well. All 74 were confirmed and none remained pending; six had already been spot-checked in Week 2 on their arXiv abstract pages. The pass found one wrong publication year. The journal version of one study (Wu et al., 2026) is dated June 2026, but the screening files had recorded 2025, the year of acceptance that appears inside its DOI; the record was corrected to 2026. Where a preprint and its published version differed in title or author list, the published version was cited.

The third concerned the numbers. A script compared the master comparison table with the extraction records, and every one of the 48 rows agreed with them in its author names, accuracy figures and cross-generator flags. For this report the wording of the table was shortened, and a second script confirmed that every number and every Yes/No flag in the version printed in Chapter 12 still equals the checked original. Every value that Chapters 9 to 13 quote from a study is taken from the extraction records.

9. TAXONOMY OF DETECTION APPROACHES

Detectors differ less in the decision they reach than in where the evidence behind it comes from, and this chapter is organised around that point. Five families result. In the first, a person designs the measurement (Section 9.1); in the second, a convolutional network learns it while being trained for detection (Section 9.2). The third trains or fine-tunes a transformer backbone (Section 9.3), the fourth reuses a frozen, general-purpose vision-language representation (Section 9.4), and the fifth contributes an explanation rather than a decision (Section 9.5). Earlier surveys sorted this work by medium or by the type of forgery (Verdoliva, 2020; Wang et al., 2023b). This review prefers sorting by the origin of the evidence because it links the taxonomy to the generalisation problem of Chapter 7: how well a detector handles a new generator depends largely on how closely its evidence is tied to its training generator. The boundaries are not watertight, and the few studies that span two families are noted where they appear. Figure 9.1 summarises the five families and their main techniques.

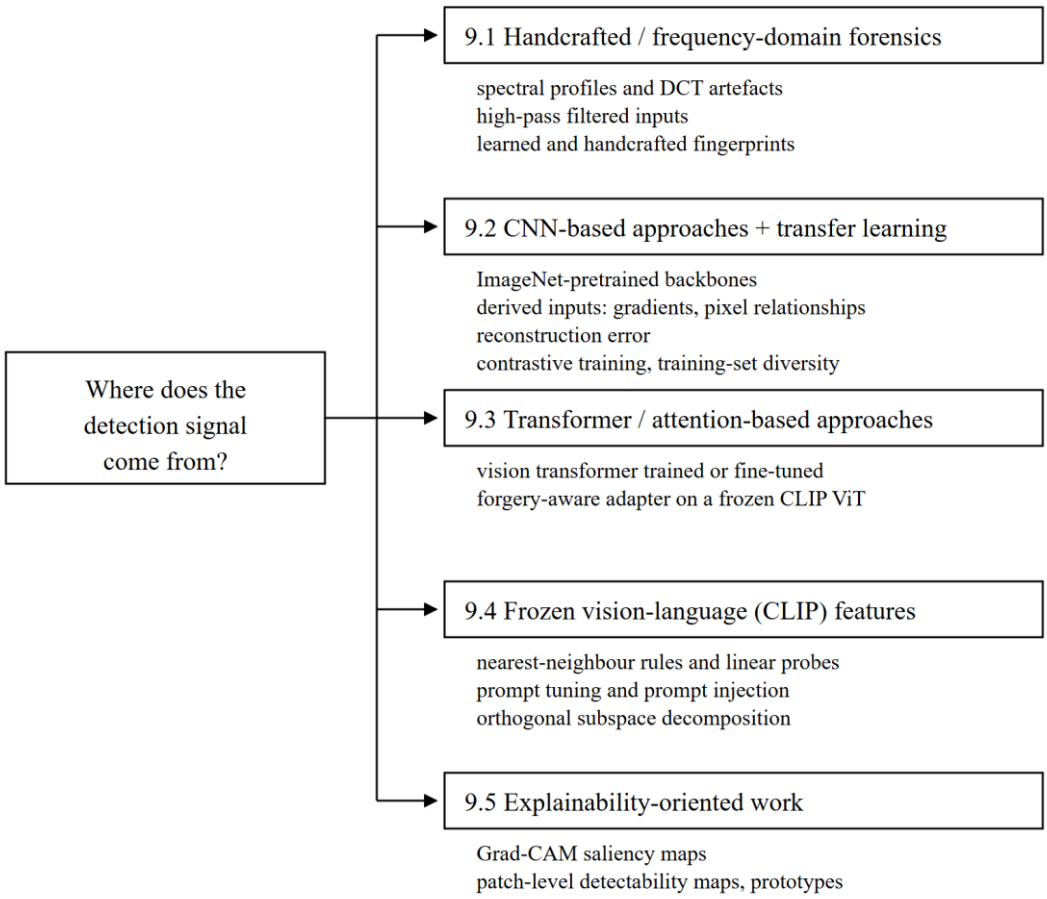


Figure 9.1. Taxonomy of the surveyed detection approaches, grouped by where the detection signal comes from. Each family carries the number of the section that discusses it and a list of representative techniques.

Table 9.1 shows how the 48 primary studies are spread across the families. CNN-based work forms the largest group and transformer-based work the smallest.

Table 9.1. Distribution of the 48 primary studies across the five families

Family	**Source of the detection signal**	**Primary studies**	**Section**
Handcrafted, statistical and frequency-domain forensics	Fixed measurements chosen by a human designer	14	9.1
CNN-based approaches and transfer learning	Features a CNN learns while being trained for detection	19	9.2
Transformer and attention-based approaches	A transformer backbone trained or fine-tuned as the detector	2	9.3
Multimodal and foundation-model-based approaches	Frozen, general-purpose vision-language features	8	9.4
Explainability in detection	Insight into why a decision was reached	4	9.5

Three studies are counted in two families, and the four studies centred on datasets and benchmarks, by Grommelt et al. (2024), Yan et al. (2025a), Zhu et al. (2023) and Rahman et al. (2023), are left out of the families and analysed in Chapters 10, 12 and 13.

9.1 Handcrafted, Statistical and Frequency-Domain Forensics

In this family a person, not an optimiser, chooses what to measure. A fixed transform, such as a Fourier or discrete cosine transform (DCT), a high-pass filter or a denoising residual, is applied to the image, and its output goes to a simple classifier, in the tradition of classical media forensics (Verdoliva, 2020). Chapter 7 explained why most of these transforms are spectral; this section reports what detectors built on them achieved.

Early spectral detectors were accurate and cheap. Durall et al. (2019) averaged the power spectrum of the discrete Fourier transform azimuthally into a one-dimensional profile, classified it with k-means, logistic regression or an SVM, and reported 100% accuracy on their own Faces-HQ set with as few as 20 labelled examples. In a follow-up, Durall et al. (2020) reached 100% with this feature on CelebA but only 85% on individual FaceForensics++ frames, and proposed suppressing the trace by regularising a generator's spectrum. Frank et al. (2020) turned to the DCT domain, in which the grid left by upsampling shows up directly. Applied to DCT coefficients, ridge regression told real and fake FFHQ images apart at 100.00%, compared with 75.78% when the same classifier was given raw pixels, although the artefact is demonstrated on the GANs they tested, not on one withheld from training. Dzanic et al. (2020) modelled how the spectrum decays at high frequencies and fed the parameters of that fit to a five-nearest-neighbour classifier, which reached 99.2% on uncompressed images of 1024×1024 pixels.

Two findings limit these results. First, the same detector drops to 88.8% on 256×256 images, and after 85% JPEG compression generated and real spectra become essentially indistinguishable (Dzanic et al., 2020). Second, Chandrasegaran et al. (2021) retrained generators with their final layer switched to bilinear upsampling, and detection based on spectral decay fell from 84.8–99.8% to 0–0.3%, so the cue stems from a design decision rather than being inherent in CNN-generated images. Zhang et al. (2019) had met a form of this earlier: their classifier learned from log spectra of

AutoGAN-simulated artefacts, not real generated images, and averaged 97.2% accuracy in leave-one-out testing but failed on GauGAN, whose upsampler differs. Since then, the frequency behaviour of generative models has been studied for its own sake (Schwarz et al., 2021).

Later methods stayed with frequency cues but let the network do more of the work. BiHPF applies bilateral high-pass filters before a ResNet-50 and, after training on four ProGAN classes, reaches a mean accuracy of 84.2% on unseen GANs (Jeong et al., 2022a). FrePGAN adds a learned frequency-level perturbation in front of that backbone and, also trained on four ProGAN classes, averages 79.4% accuracy across models; its authors openly admit that frequency artefacts are an advantage on the training generator but a hindrance to transfer (Jeong et al., 2022b). FreqNet applies its convolutions in the fast Fourier transform (FFT) domain, using a network of 1.9M parameters trained with no pretraining, and across 17 GAN test sets it reports a mean accuracy of 92.8%, whereas a baseline built on CLIP features reaches 83.0% (Ojha et al., 2023; Tan et al., 2024a). NPR replaces the spectrum with a local measurement, the relationships between neighbouring pixels that upsampling leaves behind, and reads it with a CNN of 1.44M parameters, obtaining 92.2% mean accuracy on 28 generators spanning GAN, deepfake and diffusion types (Tan et al., 2024b).

Not all handcrafted features are spectral. Nataraj et al. (2019) fed co-occurrence matrices of pixel values to a deep CNN and scored above 99% accuracy within each of two GAN datasets; they also trained on one generator family and tested on the other, although only those two families were covered. Another strand targets one specific model rather than a family. Marra et al. (2019) averaged the denoising residuals of images that share a single source, as is done to estimate camera sensor noise, and identified that source with 90.3% accuracy on uncompressed images and 90.1% after JPEG compression at quality 95. Yu et al. (2019) instead learned the fingerprint, reaching 99.43% in a five-way attribution task on CelebA (real images plus four GANs), while a PRNU-style baseline (photo-response non-uniformity) reached 86.61%. Both are closed-set: they choose among known sources rather than flagging a new one. Diffusion-era analysis of this kind exists too, with no detector attached. Corvi et al. (2023a) compare the artefacts of fifteen GAN, VQ (vector-quantised), diffusion and latent-diffusion models; they give no detection metrics and point out that post-

processing may modify the artefacts and hide them. Another contribution offers a dataset and a detector aimed directly at diffusion output (Bammey, 2024).

9.2 CNN-Based Approaches and Transfer Learning

Rather than designing the measurement, a study can let a convolutional network learn it, usually through transfer learning: a backbone pretrained on ImageNet, from families such as VGG, ResNet, DenseNet or EfficientNet (Simonyan and Zisserman, 2015; He et al., 2016; Huang et al., 2017; Tan and Le, 2019), provides general features and is fine-tuned on real and generated images. Wang et al. (2020) set the baseline for this family. Their ResNet-50 saw only ProGAN images during training, augmented with blur and JPEG, and averaged 90.8 mAP across eleven test sets, one from ProGAN itself and ten from generators it had never seen, against 90.1 without augmentation. On Photoshop-style manipulations it does no better than chance, and its authors cautioned that the transfer holds "for now".

Chai et al. (2020) narrowed the receptive field so that each decision depends on local patches alone. Their truncated Xception classifier, trained on ProGAN, achieved 99.99 AP on StyleGAN. Because each patch is scored separately, the scores also map where the evidence lies, so the study returns in Section 9.5.

Several studies then pursued universal GAN detection. Gragnaniello et al. (2021) compared seven detectors and found that on uncompressed images from unseen GANs several of them exceeded an AUC of 0.9, but compression and resizing caused severe degradation. Their own variant without downsampling went past 90% accuracy, yet the authors judged that reliable tools were still a long way off. Cozzolino et al. (2021) extended that design with contrastive training, achieving an AUC of 0.997 and 95.6% accuracy over seven unseen GAN architectures. Other studies changed where the network looks. Gram-Net relies on global texture statistics; in-domain it scores above 98%, but it manages around 77.7% when trained on StyleGAN and evaluated on DCGAN, DRAGAN and StarGAN (Liu et al., 2020). A ResNet-50 with two branches, fusing global features with patches picked out by attention, achieves a mean AP of 91.73% on the unseen models of a 19-generator set (Ju et al., 2022). An ensemble of five EfficientNet-B4 networks (Tan and Le, 2019), trained orthogonally, reaches an AUC of 0.9995 on a test set containing StyleGAN3, which was kept out of training (Mandelli et al., 2022).

These figures all came from GANs, and diffusion models erased much of the margin. Corvi et al. (2023b) tested four GAN-trained detectors on twelve generators, among them the GAN used for training and five diffusion models (ADM, GLIDE, Latent Diffusion, Stable Diffusion and DALL·E 2). Averaged over all twelve, their accuracy was roughly 52–76% on uncompressed images and about 50–61% after compression and resizing, and the diffusion models were among the hardest cases. Ricker et al. (2024b) quantified this collapse as an average AUROC drop of 15.2% and found it asymmetric: a diffusion-trained detector achieved an average Pd@1%FAR (detection rate at a 1% false-alarm rate) of 94.26% on GANs, but a GAN-trained one achieved 26.34% on diffusion images.

One remedy leaves the classifier unchanged and gives it a map computed from the image instead; NPR, described in Section 9.1, works this way (Tan et al., 2024b). LGrad turns an image into the gradients of a pretrained CNN and classifies those. In its single-class setting it averages 86.3% accuracy over eight test models, 11.4 points above FrePGAN, though it swaps reliance on the training data for reliance on the network that computes the gradients (Tan et al., 2023). With diffusion images the map is most often a reconstruction error. DIRE subtracts the DDIM reconstruction of an image from the image itself and passes the residual to a ResNet-50, averaging 99.9% accuracy across eight diffusion models (Wang et al., 2023a). SeDID uses the per-step denoising error, and its network variant averages 0.9634 AUC across the three test sets it reports (Ma et al., 2023). LaRE² measures the error in latent space and uses it to refine features, improving accuracy by up to 11.9% over the previous best on eight GenImage generators (Luo et al., 2024). AEROBLADE does away with the classifier and scores an image by how faithfully the latent diffusion model's own autoencoder can rebuild it, giving a mean AP of 0.992 but requiring access to that autoencoder (Ricker et al., 2024a). FakeInversion inverts Stable Diffusion to obtain a noise map and a reconstruction for each image, and a ResNet-50 trained from scratch takes all three as input, averaging 0.759 AUC-ROC over text-to-image models unseen in training (Cazenavette et al., 2024).

The last group alters the training signal and leaves the input alone. DRCT turns real images into hard positives by reconstructing them with a diffusion model, then trains contrastively on these, gaining over 10% accuracy in cross-set tests, though the gain is smaller on GAN images (Chen et al., 2024). ZED uses no generated training

images: it fits a lossless coding model to real images and flags those it can encode unusually cheaply, averaging about 90.0% AUC over 19 generators, but is unreliable when inputs are heavily compressed or resized (Cozzolino et al., 2024b). Community Forensics attacks the problem through data. Trained on 2.7M images produced by 4,803 generators, it reaches 89.2% accuracy and 0.987 mAP on an out-of-distribution set of 21 models, which in its authors' view still leaves error rates too high for critical applications (Park and Owens, 2025).

Overall, the family is moving away from convolutional networks trained from scratch to find forensic features and towards general-purpose pretrained representations, kept frozen in some detectors and fine-tuned in others. DRCT already relies on CLIP-ViT-L/14 as one of its backbones (Chen et al., 2024), and Community Forensics classifies with a CLIP-pretrained ViT-S (Park and Owens, 2025); such representations are the subject of Sections 9.3 and 9.4.

9.3 Transformer and Attention-Based Approaches

In a vision transformer (ViT), an image is handled as a sequence. It is divided into patches of a fixed size, each patch is turned into a vector by a linear embedding, and self-attention lets every patch draw on all the others from the first layer onwards (Dosovitskiy et al., 2021). A convolutional network works the other way round, widening its view of the image only gradually as layers of local filters are stacked. For detection, the contrast has two sides. Most of the evidence described in Sections 9.1 and 9.2 is local and high-frequency. Several of the better convolutional detectors work by preserving that fine detail, one by letting each decision see only a small patch of the image (Chai et al., 2020) and others by dropping the downsampling in the first layer, which would otherwise remove that detail (Cozzolino et al., 2021; Gragnaniello et al., 2021). A transformer's patch embedding is a stride-16 projection, so the model starts by discarding part of the information those detectors are careful to retain. In exchange, its representation is built from relations spanning the entire image, not from filter responses within a local window.

Only one surveyed study isolates the architecture. Coccomini et al. (2022) compared a ViT-Base with an EfficientNetV2-M, training both on subsets of ForgeryNet and testing them on all fifteen of its manipulation methods. The CNN turned out to be the more specialised of the two. In one experiment the training data

held three face-transfer methods: FaceShifter, FS-GAN and Deepfakes. The CNN reached 81.1% on Deepfakes but only 47.0% on FaceShifter and 57.0% on FS-GAN, although all three were in its training set, while the ViT stayed close to 62.0% on each. Taking their experiments together, the authors conclude that the CNN often scores higher on the methods it was trained on but generalises worse to new ones, and that the ViT's accuracy varies less from one method to the next. They also admit that accuracy on new methods remains low for both models.

The family's second primary study adapts an existing transformer instead of training one for detection. FatFormer retains a CLIP ViT and adds a forgery-aware adapter that operates jointly on image and frequency representations, which the text side of the model helps to align (Liu et al., 2024a). With the standard four-class ProGAN set for training, it reaches 98% accuracy on GANs and 95% on diffusion models, in both cases on generators that were absent from its training data. Because FatFormer's representation is drawn from a frozen vision-language model, Section 9.4 discusses it again. The point for this section is that the useful transformers in this literature are almost never trained from random initialisation.

Studies covered elsewhere in this review bear this out. DRCT, discussed in Section 9.2 for its contrastive training, has both ConvNeXt-B and CLIP-ViT-L/14 backbones (Chen et al., 2024). The Community Forensics detector, likewise discussed in Section 9.2, classifies with a CLIP-pretrained ViT-S and offers ConvNeXt-S as an alternative (Park and Owens, 2025). The prototype method of Section 9.5 feeds the features of a fine-tuned ViT to an SVM (Aghasanli et al., 2023). One secondary study also trains contrastively, with global and local similarities, on a large diffusion-image collection that it releases (Baraldi et al., 2024).

None of these comparisons isolates the backbone, however. In this literature a change of backbone usually brings different pretraining data, a different training objective and a different amount of supervision, all at once. The studies therefore demonstrate that transformer-based systems did well; they do not demonstrate that the transformer is responsible. Only Coccomini et al. (2022) hold everything else constant. This is one reason why, in the second phase of this project, the ViT is planned as a benchmark for comparison with CNN backbones trained under matched conditions, not as a proposed improvement.

9.4 Multimodal and Foundation-Model-Based Approaches

Strictly speaking, CLIP was never built to detect anything. It pairs an image encoder with a text encoder, trained together on captioned images so that each image lands near its own caption, and the visual representation this produces carries over to tasks it was not trained on (Radford et al., 2021). The detectors in this section rest on an empirical finding: real and generated images are separable in that representation, although forensics played no part in its training.

Ojha et al. (2023) first established this. They kept the weights of CLIP's ViT-L/14 image encoder fixed and only read off its features, which were classified either by a nearest-neighbour rule or by a linear probe fitted to the widely used ProGAN training set. Compared with detectors that learn their own backbone from the same data, the frozen approach improved mAP by 15.07 and accuracy by 25.90% on diffusion and autoregressive models not seen in training. Why the frozen features behave this way is, the authors admit, still an open question, and later work accepts the representation and looks for the best way to read it.

Several strategies have been tried. The simplest fits a small classifier to the frozen features: Cozzolino et al. (2024a) fit a linear SVM to next-to-last-layer CLIP features using a reference set that can be as small as ten paired images from a single generator, and obtain an average AUC of about 90.0% over eighteen generators, with AUC gains of 6% on out-of-distribution data and 13% on impaired images. Another reads more of the network: RINE draws on the intermediate encoder blocks, not only the final one, and learns how much weight each deserves; with four-class training it records an average accuracy of 91.5% and an AP of 98.8 across twenty test sets (Koutlis and Papadopoulos, 2024).

A third attaches trainable components to the frozen model. Examples are FatFormer's adapter, whose 98% and 95% accuracies on unseen GANs and diffusion models were given in Section 9.3 (Liu et al., 2024a), and prompt tuning, which Khan and Dang-Nguyen (2024) compare with adapters, linear probing and full fine-tuning; with it they obtain an mAP of 98.06% and an average accuracy of 89.45% over eighteen unseen generators. C2P-CLIP turns to the text side instead, where a category-common prompt injected into its text encoder carries detection semantics into the visual features. On the nineteen subsets of UniversalFakeDetect it reaches an accuracy of 93.79% with

an mAP of 98.66%, and on GenImage its accuracy is 95.8% (Tan et al., 2025). Effort takes a geometric view, running a singular value decomposition (SVD) on the frozen features and operating within orthogonal subspaces derived from it, for an mAP of 99.41% and a mean accuracy of 95.19% on those nineteen subsets (Yan et al., 2025b).

In other work the text prompt is an input rather than part of the mechanism. DE-FAKE, aimed at text-to-image generation, sets an image-only ResNet18 against a hybrid classifier over both CLIP encoders. After training on Stable Diffusion, the hybrid achieves accuracies of 0.932 on Latent Diffusion, 0.899 on GLIDE and 0.885 on DALL·E 2, while the authors judge image-only detection on these models to fall well short of what they designed it to achieve (Sha et al., 2023). Further secondary studies extend these ideas to prompt-tuned vision-language models that serve directly as detectors (Chang et al., 2023), language-guided contrastive training (Wu et al., 2026), mixtures of low-rank experts (Liu et al., 2024b), diffusion-image detection with such features (Coccomini et al., 2024) and a captioning reformulation of the task (Keita et al., 2025).

This line of work has two gaps. The first is interpretability: the paper that started the approach and the few-shot method above both note that what CLIP features capture in forensic terms is not understood (Ojha et al., 2023; Cozzolino et al., 2024a), a problem Section 9.5 takes up. The other gap is robustness. Under a combination of blur, cropping, compression and noise, RINE degrades considerably (Koutlis and Papadopoulos, 2024), and with prompt tuning, accuracy on images from commercial tools drops once they are JPEG-compressed (Khan and Dang-Nguyen, 2024). The evaluation protocol barely changes from study to study either. Several of the detectors discussed here are fitted to ProGAN images but reported on diffusion test sets (Ojha et al., 2023; Khan and Dang-Nguyen, 2024; Koutlis and Papadopoulos, 2024; Liu et al., 2024a; Yan et al., 2025b). Their headline generalisation figures therefore capture one particular shift between training and test data rather than the general case.

9.5 Explainability in Detection

A probability on its own gives no reason for a detector's decision. For the forensic uses described in Chapter 1 this is a genuine limitation, so a few studies ask where in the image a decision comes from. Grad-CAM, the standard tool, takes the feature maps of one convolutional layer, weights each map by how strongly the class

score responds to it (its gradient), and sums the weighted maps into a coarse heatmap over the image (Selvaraju et al., 2017). A later variant, Grad-CAM++, improves the weighting for sharper localisation, especially when an image contains several instances of the same class (Chattopadhyay et al., 2018). Both methods come from object recognition, where the highlighted area contains something visible; forensic evidence, as Chapter 7 showed, often is not.

The mismatch is easy to see in the CIFAKE study. Bird and Lotfi (2024) built a 120,000-image dataset of their own: 60,000 real photographs from CIFAR-10 and 60,000 images produced by Stable Diffusion v1.4. On it they trained a small custom CNN, chosen from 36 candidate topologies, with two 32-filter convolutional layers and a 64-unit dense layer; it reached 92.98% accuracy. The authors report that, according to Grad-CAM, the network's classifications rest on small background imperfections rather than on the subject. The explanation is valuable exactly because it does not flatter the detector: what the network picked up belongs to a single generator at a 32-pixel resolution, not to synthesis in general.

Chai et al. (2020) approach localisation from the opposite side. Their patch classifiers, covered in Section 9.2, confine each decision to a small region by limiting the receptive field, and combining the per-patch scores yields a map of the image regions that are detectable at all. Such a map explains the evidence, not a particular model's reasoning, and one cannot stand in for the other.

Aghasanli et al. (2023) prefer interpretability by construction to explanation after the fact, classifying the features of a fine-tuned ViT with an SVM whose support vectors serve as prototypes to which any decision can be traced back. Accuracy within one dataset is high: with the fine-tuned ViT followed by an MLP head they report 99.7% on a class-conditional ImageNet set, 96.7% on an image-to-image ImageNet set, 99.5% on Oxford-IIIT Pet and 99.4% on FFHQ, and the SVM on the same features reaches 93.0–97.2%. Their cross-dataset tests give lower figures, 81.3% and 84.0%. These tests say nothing about unseen generators: the fakes in both the training and the test data came from Stable Diffusion, so the image content changed but the generator did not. The authors also mention having to build the datasets themselves, as no open ones were available.

Only recently have such explanations been evaluated quantitatively. Tsigos et al. (2024) put forward an evaluation framework and test it on an EfficientNetV2 detector with FaceForensics++. Grad-CAM++, RISE, SHAP, LIME and SOBOL are compared on one common measure: how far the detector's accuracy falls once the image segments that each method ranks highest have been altered; the lower the accuracy that remains, the better the explanation. LIME comes out on top: altering only its highest-ranked segment left the detector with an accuracy of 0.579 on NeuralTextures, and altering its three highest-ranked segments brought accuracy on DeepFakes down to 0.245. The type of forgery also matters: the authors find Face2Face and FaceSwap fakes the hardest to explain. The base is narrow, one detector on one dataset, but the study shows that explanation quality can be measured rather than left to inspection.

The open problem is the step from showing where a network looked to stating what it found there. A heatmap can reveal a shortcut, as with CIFAKE, yet it leaves open which property of the underlying pixels is the synthetic one. The frozen-feature detectors of Section 9.4 are further still from an answer, since their authors acknowledge not knowing what the representation encodes (Ojha et al., 2023; Cozzolino et al., 2024a). CIFAKE also links this chapter to the next. Its contribution combines a dataset with an explainability analysis (Bird and Lotfi, 2024), and Chapter 10 turns to the datasets and benchmarks on which the field relies.

10. DATASETS AND BENCHMARKS

Reported accuracy figures can only be read in the light of the data behind them, so this chapter looks at the datasets and benchmarks that the surveyed detectors are trained and tested on. Section 10.1 traces how these datasets developed, from the first GAN benchmarks to in-the-wild collections, and Section 10.2 sets out the problems known to affect them.

10.1 From GAN Benchmarks to In-the-Wild Data

The field's evaluation practice was set by ForenSynths, the dataset that Wang et al. (2020) released with their CNN detector. Its lasting value lies in the evaluation design more than in the images. Training uses one generator only: 720,000 images from 20 ProGAN categories, with 18,000 real and 18,000 generated images in each. Performance is then reported separately on eleven test sets: one from ProGAN itself and ten from generators that the detector never sees in training. This design made cross-generator transfer, that is, how well a detector handles generators missing from its training data, a headline figure instead of an afterthought.

Several later detectors are trained on this ProGAN set: some on one, two or four of its twenty categories (Jeong et al., 2022a; Tan et al., 2023; Koutlis and Papadopoulos, 2024; Liu et al., 2024a; Tan et al., 2024a; Tan et al., 2024b), one on 200,000 of its images (Khan and Dang-Nguyen, 2024) and one on all twenty categories (Yan et al., 2025b). The price of this agreement is that much of the literature trains on a single distribution, so the generalisation results placed side by side all describe one particular step from training data to test data, not the general case. Ojha et al. (2023) widened the design on the test side, keeping the ProGAN training set and adding test subsets drawn from diffusion models and from autoregressive models (which build an image one part at a time). The resulting benchmark, UniversalFakeDetect, has nineteen test subsets, now the standard test suite for the CLIP-feature detectors described in Section 9.4 (Ojha et al., 2023; Tan et al., 2025; Yan et al., 2025b).

Diffusion models called for new data. DiffusionForensics, published with the DIRE detector, sets real images from LSUN-Bedroom and ImageNet against images from eight diffusion models (Wang et al., 2023a). GenImage is larger: it pairs 1,331,167 real ImageNet images with over a million generated ones from eight generators

(Midjourney, two versions of Stable Diffusion, ADM, GLIDE, Wukong, VQDM and BigGAN). More importantly, it makes cross-generator classification and the classification of degraded images, such as low-resolution, JPEG-compressed or blurred ones, benchmark tasks in their own right (Zhu et al., 2023). ArtiFact covers more generators, 25 generation methods across 2,496,738 images, and was designed with robustness as an aim (Rahman et al., 2023). CIFAKE sits at the other end of the scale: its 120,000 images pair photographs from CIFAR-10 with Stable Diffusion v1.4 images at a resolution of 32×32 (Bird and Lotfi, 2024). Its small size allows training on modest hardware, which is both its attraction and its limitation. D3 is a further diffusion-era collection, released together with a detector trained contrastively (Baraldi et al., 2024).

The third generation of datasets stops relying on a hand-picked list of generators. Community Forensics takes diversity to its limit, collecting 2.7 million images made by 4,803 different generators to test whether a broader range of training data is enough by itself to produce generalisation (Park and Owens, 2025). Chameleon heads the other way, gathering realistic images and reporting how accurately detectors classify them (Yan et al., 2025a). TWIGMA's images were collected from Twitter, so the dataset reflects what users actually shared, not what a researcher decided to generate (Chen and Zou, 2023). Table 10.1 lists these datasets by year of introduction, with each one's scale, source of real images, generators covered and notes, including the limitation its authors state. It also includes WildFake, a hierarchical dataset of AI-generated images (Hong et al., 2025). WildFake, D3 and TWIGMA entered the review only as secondary studies, without full-text extraction, so the table leaves some of their properties unstated.

Table 10.1. Datasets and benchmarks that the surveyed studies use to train and test their detectors, in order of the year each was introduced

Dataset	Introduced by	Year	Scale	Real-image source	Generators covered	Notes / stated limitation
ForenSynths (CNNDetection)	Wang et al. (2020)	2020	720K training images, 18,000 fake and 18,000 real in each of 20 ProGAN categories (Tan et al., 2023); 11 per-generator test sets	LSUN	Training: ProGAN; 11 test sets: BigGAN, CRN, CycleGAN, DeepFakes, GauGAN, IMLE, ProGAN, SAN, SITD (StyleGAN2 tested separately), StarGAN, StyleGAN	GAN-era protocol: ProGAN for training, unseen generators for testing. Standard training set in later work (Tan et al., 2023; Tan et al., 2024a; Tan et al., 2024b); several more detectors train on ProGAN, using 1, 2 or 4 of the 20 categories (Jeong et al., 2022a; Koutlis and Papadopoulos, 2024; Liu et al., 2024a), 200K of its images (Khan and Dang-Nguyen, 2024) or all 20 categories (Yan et al., 2025b). The authors' "for now" caveat: future generators may escape detection
UniversalFakeDetect	Ojha et al. (2023)	2023	720K ProGAN images for training; 19 test subsets (Tan et al., 2025; Yan et al., 2025b)	—; training set reused from ForenSynths (Wang et al., 2020)	BigGAN, CRN, CycleGAN, DALL-E, DeepFakes, GauGAN, Glide, Guided, IMLE, LDM, ProGAN, SAN, SITD, StarGAN, StyleGAN	Adds diffusion and autoregressive subsets to the test side of the ForenSynths protocol; the 19 subsets have since become the standard test suite for CLIP-feature detectors (Tan et al., 2025; Yan et al., 2025b)
DiffusionForensics	Wang et al. (2023a)	2023	—	LSUN-Bedroom, ImageNet	8 diffusion models: ADM, DDPM, iDDPM, LDM, PNDM, SD-v1, SD-v2, VQ-Diffusion	Published together with the DIRE detector; covers diffusion-generated images only
GenImage	Zhu et al. (2023)	2023	More than 1M fake and 1,331,167 real images	ImageNet	8 generators: ADM, BigGAN, GLIDE, Midjourney, SD v1.4, SD v1.5, VQDM, Wukong	Sets out two explicit tasks: cross-generator and degraded-image classification. Its authors report that detection drops sharply across generators and for low-resolution, JPEG-compressed and blurred inputs. Later shown to carry biases in JPEG compression and image size (Grommelt et al., 2024)
ArtiFact	Rahman et al. (2023)	2023	2,496,738 images in total (964,989 real and 1,531,749 fake)	—	13 GANs, 7 diffusion and 5 other methods (25 in all)	Aimed at robustness; the authors' own detector was scored on the IEEE VIP Cup 2022 test sets. The authors note that state-of-the-art detectors still have difficulty with generators absent from training
TWIGMA	Chen and Zou (2023)	2023	—	Not applicable: AI-generated images and their metadata, collected from Twitter	—	Gathered from a platform, not generated for research; included at screening as evidence of synthetic images already circulating in the wild

Dataset	Introduced by	Year	Scale	Real-image source	Generators covered	Notes / stated limitation
CIFAKE	Bird and Lotfi (2024)	2024	60,000 real and 60,000 synthetic images, 120,000 in total	CIFAR-10	Stable Diffusion v1.4, a latent diffusion model	One generator only, at a 32×32 resolution; the authors' Grad-CAM analysis shows background imperfections driving the classification
D3	Baraldi et al. (2024)	2024	—	—	—	Large set of diffusion images, published at ECCV 2024 alongside a contrastive detection method; included at screening as a secondary study, hence no full-text extraction row
Chameleon	Yan et al. (2025a)	2025	—	—	—	A benchmark of realistic in-the-wild images. AIDE, the detector proposed in the same study, averages 92.77% accuracy on AIGCDetectBenchmark and 86.88% on GenImage but reaches only 65.77% on Chameleon; the authors describe the problem as far from solved
Community Forensics	Park and Owens (2025)	2025	2.7M images made by 4,803 generators	—	4,803 generators, covering GANs, commercial models and both latent and pixel diffusion	Made to check whether diversity of generators leads to generalisation. Its authors note that diffusion models are over-represented in the dataset and that error rates are still too high for critical uses
WildFake	Hong et al. (2025)	2025	—	—	—	Hierarchically organised set of AI-generated images; a secondary study included for this table alone, hence no full-text extraction row

Note: a dash (—) means that the review's extraction records give no value for that property, not that the dataset lacks it. Text without a citation draws on the study named in the "Introduced by" column; a citation anywhere else in a row points to a different study.

10.2 Known Problems with the Evaluation Data

Of the weaknesses a dataset can have, the most direct is a shortcut: a property that separates its real images from its generated ones without being a trace of generation. Grommelt et al. (2024) found two such properties in GenImage. Its real images and its generated ones differ systematically in two respects, JPEG compression and image size, which lets a classifier score highly by reading these properties rather than any forensic trace. Once the bias is removed, in-domain accuracy (accuracy on test images from the training generator) falls, but the result that matters improves: on GenImage's cross-generator task, accuracy went up by more than eleven points, with a ResNet50 improving from 71.68% to 82.74% and a Swin-T from 74.09% to 85.83%. The authors caution that other unwanted biases, such as differences in style and generator fingerprints (Section 7.3), may remain even with these two controlled.

Single-generator datasets show a narrower form of the same weakness. CIFAKE is built from exactly one generator, and its authors' Grad-CAM analysis, which highlights the image regions behind the network's decision (Section 9.5), reports that the trained network relies on small imperfections in the backgrounds of the images (Bird and Lotfi, 2024). These imperfections belong to Stable Diffusion v1.4 at a 32×32 resolution, not to synthesis in general, so accuracy on CIFAKE says little about another generator. GenImage measures the gap under matched conditions: a ResNet-50 has an accuracy of 99.9% when trained and tested on images from one generator but only 54.9% on Midjourney images (Zhu et al., 2023). A single-generator dataset cannot reveal such a fall, which is why the study planned for Semester 8 (Section 14.3) pairs CIFAKE with a subset of GenImage instead of using CIFAKE alone.

Staleness, meaning that a dataset falls behind the generators released after it, has the same root. Each dataset's generator list is frozen at publication, and the field has responded by lengthening the list: one recent evaluation covers nineteen generators (Cozzolino et al., 2024b), another eighteen (Cozzolino et al., 2024a), and Community Forensics 4,803 (Park and Owens, 2025). Longer lists keep the benchmarks moving, yet they never quite catch up with whichever generator appeared most recently. Comparability is also harmed when researchers create their own data. Several of the surveyed studies test on sets assembled for their own paper (Durall et al., 2019; Durall et al., 2020; Ju et al., 2022; Aghasanli et al., 2023), and Aghasanli et al. (2023) give the reason directly: no open datasets existed for their setting.

The strongest evidence that benchmark scores overstate real-world performance comes from Chameleon. In the study that introduced it, the authors' own detector reaches a mean accuracy of 92.77% on a standard benchmark, AIGCDetectBenchmark, and 86.88% on GenImage, but only 65.77% on the realistic images collected for Chameleon; every detector they tested comes close to failing there, and they judge the problem to be far from solved (Yan et al., 2025a). The benchmark figure is thus twenty-seven points above what a detector achieves on images like those it would face in practice. The gap lies in the data, not in any one detector, which is why Chapter 11 turns to the measurement of performance and to how the surveyed studies report it.

11. EVALUATION METRICS

11.1 Metrics Used in Detection

At test time, every detector reviewed here acts as a binary classifier labelling each image real or generated. By convention in this literature, the AI-generated image is the positive class, so every prediction falls in one of four cells. A generated image labelled generated is a true positive (TP), while one passed as real is a false negative (FN). A real image labelled real is a true negative (TN), while one flagged as generated is a false positive (FP). Each metric below condenses these four counts differently; Figure 11.1 shows them as a confusion matrix with the derived metrics.

Confusion matrix (positive class: AI-generated image)

	Predicted: generated	Predicted: real
Actual: generated	TP true positive	FN false negative
Actual: real	FP false positive	TN true negative

Metrics derived from the four counts

Accuracy = (TP + TN) / (TP + FP + FN + TN)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1 = 2 · (Precision · Recall) / (Precision + Recall)

AUC (area under the ROC curve) and AP (average precision) summarise performance over a sweep of decision thresholds, not at one fixed point.

Figure 11.1. Confusion matrix for real-or-generated classification (generated images as the positive class) and the measures computed from it. AUC and AP are threshold-free, summarising behaviour over all decision thresholds rather than one operating point.

Accuracy is the fraction of all predictions that are correct:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{11.1}$$

No measure is reported more often in this field. Its popularity can be defended only because benchmark datasets are usually balanced by design; CIFAKE, for

example, holds 60,000 of each class (Bird and Lotfi, 2024). On an imbalanced test set the same value would mostly reflect the class prior, the proportion of each class.

Precision and recall separate the two kinds of error. Precision is the share of images flagged as generated that really are generated, so false positives lower it. Recall, or the true positive rate (TPR), is the share of generated images that are caught, so false negatives lower it:

$$Precision = \frac{TP}{TP + FP} \tag{11.2}$$

$$Recall = TPR = \frac{TP}{TP + FN} \tag{11.3}$$

The two pull in opposite directions; the F1 score merges them into their harmonic mean, an average that is high only when both are high:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{11.4}$$

These four metrics all depend on the decision threshold, the score above which an image is called generated. The remaining measures avoid fixing one threshold by sweeping it through its entire range. The first of them also uses the false positive rate (FPR), the share of real images flagged as generated:

$$FPR = \frac{FP}{FP + TN} \tag{11.5}$$

Plotting TPR against FPR across the sweep gives the receiver operating characteristic (ROC) curve, and the area under that curve is the AUC:

$$AUC = \int_0^1 TPR(FPR)\, d(FPR) \tag{11.6}$$

An AUC of 0.5 matches random guessing, and 1.0 means perfect separation. Average precision (AP) extends the same reasoning to the precision–recall curve, weighting the precision reached at each recall level by the gain in recall since the level before, then summing:

$$AP = \sum_k (Recall_k - Recall_{k-1}) \cdot Precision_k \tag{11.7}$$

When results come from N separate test sets, usually one per generator, their AP values are averaged into the mean average precision (mAP):

$$mAP = \frac{1}{N} \sum_i AP_i \tag{11.8}$$

11.2 How the Surveyed Studies Report Results

Several specialised reporting conventions recur. The most frequent, mAP over per-generator test sets, dates from the ForenSynths protocol, under which Wang et al. (2020) report 90.8 mAP averaged across eleven test generators, ten of them unseen in training. Another convention pairs accuracy with AUC, setting a threshold-dependent figure beside a threshold-free one; Cozzolino et al. (2021), for example, report 95.6% accuracy with 0.997 AUC on seven GAN architectures unseen in training. Patch-level and single-transfer results often appear as AP alone, for instance 99.99 AP for a ProGAN-trained detector tested on StyleGAN (Chai et al., 2020). The option closest to operational practice fixes the false alarm rate and reports the probability of detection (Pd) at that rate. At a 1% false alarm rate (Pd@1%FAR), Ricker et al. (2024b) report that a diffusion-trained detector averages 94.26% detection on GAN images, while the reverse direction yields 26.34%.

The trouble lies in that variety. Studies differ in both their metrics and their test sets: mAP on eleven per-generator test sets (Wang et al., 2020), accuracy with AP on eight diffusion models (Wang et al., 2023a), an averaged AUC-ROC on unseen text-to-image models (Cazenavette et al., 2024), AUC on nineteen generators (Cozzolino et al., 2024b), and mAP plus accuracy on an evaluation set of twenty-one models (Park and Owens, 2025). Because the test sets differ, two headline figures rarely describe the same quantity, and no reader can convert one metric into another afterwards.

Thresholds complicate matters further. Precision, recall, F1 and accuracy all rest on one operating point, usually the default, which is almost never recalibrated between datasets; any accuracy figure therefore hides a threshold choice. AUC and AP escape that choice, yet are silent on performance at the threshold a deployed system would really use. The exception is Pd@1%FAR (Ricker et al., 2024b), which shows what the other conventions are missing.

The reported averages conceal as much as they reveal: a high mean can sit beside a complete failure on a generator that was never averaged in. A spectrum classifier from Zhang et al. (2019) reaches 97.2% average accuracy in leave-one-out testing, but it fails on GauGAN. On GenImage, a ResNet-50 scoring 99.9% on its own training generator manages only 54.9% on images from Midjourney (Zhu et al., 2023). In forensic use the generator that matters is the worst case, not the average. The master

comparison table in Chapter 12 therefore pairs every figure with its metric, its test suite and its cross-generator status.

12. COMPARATIVE ANALYSIS AND FINDINGS

Chapters 9 to 11 discussed detection approaches, datasets and metrics separately. This chapter compares the surveyed studies directly, using Table 12.1, which gives each of the 48 primary studies one row in order of publication. A row records the approach, model, data, generators covered and headline result, plus two further entries: whether the detector was tested on generators held out of training, and the limitation that its authors acknowledge. Those two final columns are this review's addition to the usual comparison. Every count below comes from tallying one column of Table 12.1 and can be verified there.

Table 12.1. The 48 primary studies compared side by side, arranged by year of publication and, within a year, by first author

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Durall et al. (2019)	Spectral forensics: power spectrum averaged azimuthally to 1D, then a classical classifier	RBF-kernel SVM, logistic regression, unsupervised K-means	Faces-HQ (authors' own, 40K), CelebA, FaceForensics++	Self-trained DCGAN, thispersondoesnotexist.com fakes, FF++ manipulations	Faces-HQ: 100% accuracy from as few as 20 labelled examples; FF++ videos: 91% accuracy	No – each dataset tested alone	Low-resolution content is harder to flag: much less of the frequency spectrum is available
Marra et al. (2019)	Handcrafted PRNU-like fingerprint from averaged denoising residuals; the source is found by correlation	Averaged denoising-filter residuals, no trained classifier	Per source 1,000 RGB images (256x256); RAISE camera photos; category sets for CycleGAN / ProGAN / StarGAN	9 CycleGAN configurations, 6 ProGAN, 5 StarGAN	90.3% source-attribution accuracy uncompressed; 90.1% at JPEG QF=95	No – known GANs only (source ID)	Poorer for StarGAN male/smiling networks and real cameras (PRNU patterns of low energy)
Nataraj et al. (2019)	Deep CNN applied to pixel co-occurrence matrices	Deep CNN fed 3-channel co-occurrence matrices	56,000+ images in two GAN datasets (unpaired translation; facial attributes)	CycleGAN, StarGAN	Over 99% accuracy on each dataset	Yes – train on one GAN, test on the other	Just two GAN families tested; the abstract states no explicit limitation
Yu et al. (2019)	Learns GAN fingerprints for fake detection and closed-set attribution	Deep CNN attribution model; fingerprint drawn from its last 1x1x512 feature/weight tensors; AutoEncoder variant visualises it	LSUN bedroom and CelebA (20,000 reals apiece); per source 100K train, 10K test	ProGAN, SNGAN, CramerGAN, MMDGAN (128x128x3 each)	5-class attribution (real + 4 GANs): CelebA 99.43%, LSUN 98.58%; 86.61 / 67.84 for a PRNU baseline	No – attributes only to known GANs	Requires labelled samples from each possible source GAN; tested at 128x128 only
Zhang et al. (2019)	Spectrum classifier aimed at GAN upsampling artefacts; an AutoGAN simulator creates them, so no real fakes are needed	ImageNet-pretrained ResNet-34 on the log frequency spectrum, phase dropped	14-category CycleGAN image sets; AutoGAN trained on 4,000 MSCOCO images	CycleGAN mainly; StarGAN, GauGAN tested too	AutoGAN-trained spectrum classifier, never shown a real fake: 97.2% mean leave-one-out accuracy	Yes – simulated artefacts expose unseen GANs	Every classifier fails on GauGAN, which upsamples differently; hurt by upsampler mismatch

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Chai et al. (2020)	Classifies patches with restricted receptive fields; heatmaps locate the detectable patches	Patch models from truncated Xception and ResNet (best: Xception blocks)	CelebA-HQ (24,183) and FFHQ (60,000) for training; FaceForensics++	Glow, GMM face generator, ProGAN, StyleGAN, StyleGAN2; FF++ manipulations	Train ProGAN / test StyleGAN: AP 99.99; unseen resolutions: AP ~100	Yes – ProGAN-trained, scored on StyleGAN/Glow/ GMM	"No method is completely bulletproof": stronger generators, out-of-distribution inputs or adversarial attacks can beat it
Durall et al. (2020)	Classical classifiers on a 1D azimuthal integral of the DFT power spectrum; spectral regularisation is also proposed	SVM, logistic regression and unsupervised K-means on that spectral feature	Faces-HQ (40K, authors' own), CelebA (202,599), FaceForensics++	CelebA: DCGAN, DRAGAN, LSGAN, WGAN-GP; FaceForensics++ deepfakes	Faces-HQ 100% (SVM, 1000 samples); CelebA 100% (2000 samples); FF++ single frames 85%	No – (un)supervised tests within each dataset only	Assumes generators that use up-convolution; with other synthesis methods these signatures may be absent
Dzanic et al. (2020)	Analyses high-frequency Fourier spectral decay via fitted parameters	KNN (k=5) using the fitted decay parameters	~5,000 FFHQ and Cat Head Detection dataset images; 10/90 train/test split	ALAE, PGGAN, StyleGAN, StyleGAN2, VQ-VAE2	Mixed-generator training: 99.2% accuracy on 1024x1024 uncompressed images	No – none of the five generators held out	85% JPEG compression leaves generated and real spectra essentially indistinguishable; at 256x256, accuracy is 88.8%
Frank et al. (2020)	DCT-spectrum analysis of grid-like artefacts left by upsampling, with simple classifiers	On DCT spectra: a shallow CNN (4-conv-layer, ~170K params) and ridge regression	LSUN bedrooms, FFHQ, CelebA	BigGAN, CramerGAN, MMDGAN, ProGAN, SN-DCGAN, StyleGAN	FFHQ real-vs-fake: Ridge-Regression-DCT 100.00% (75.78% on raw pixels); CNN-DCT source ID: LSUN 99.64%, CelebA 99.07%	No – no experiment with a held-out GAN	Purpose-built adversarial perturbations can still fool the classifier
Liu et al. (2020)	Detects fake faces from global (Gram) texture statistics	Gram-Net, a CNN built on global texture representations	Reals: CelebA-HQ, FFHQ, CelebA; GAN face sets; natural: ImageNet / BigGAN	BigGAN, DCGAN, DRAGAN, PGGAN, StarGAN, StyleGAN	>98% in-domain; ~77.7% accuracy across GANs (StyleGAN train, DCGAN / DRAGAN / StarGAN test)	Yes – tested on unseen GAN models	Studies texture contrast alone; real/fake differences are broader

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Wang et al. (2020)	Two-class real-vs-fake CNN; Blur+JPEG augmentation and careful pre- and post-processing	ImageNet-pretrained ResNet-50	ForenSynths: 720K ProGAN training images, LSUN reals, 11-generator test sets	Train: ProGAN; 11 test sets: BigGAN, CRN, CycleGAN, DeepFakes, GauGAN, IMLE, ProGAN, SAN, SITD, StarGAN, StyleGAN; StyleGAN2 tested separately	Averaged across 11 test sets (ProGAN and 10 unseen generators): mAP 90.1 without augmentation, 90.8 with Blur+JPEG(0.5)	Yes – trained on ProGAN; 10 of the 11 test generators unseen	Chance-level on traditional (Photoshop-style) edits; the authors hedge with "for now"
Chandrasegaran et al. (2021)	Counter-study of spectral-decay features: the upsampling choice can remove the high-frequency artefact	KNN (k=5) over 3 spectral features, alongside SVM and MLP; GANs retrained with altered last-layer upsampling	CelebA-HQ (512); CelebA (128) and LSUN Bedrooms (128)	DCGAN, LSGAN, StarGAN, WGAN-GP; supplementary: CRN, IMLE	Detection rate drops to 0–0.3% under bilinear upsampling (baseline 84.8–99.8%)	No – one classifier per GAN	Authors' own finding: CNN-generated images need not show high-frequency spectral decay, so detectors relying on it are unreliable
Cozzolino et al. (2021)	Contrastive learning on a network that limits sub-sampling	ImageNet-pretrained ResNet50 without first-layer subsampling, plus 2 projection layers	Train: 362K real LSUN + 362K ProGAN images; test: ImageNet, COCO, RAISE	Train: ProGAN; unseen test: BigGAN, CycleGAN, GauGAN, RelGAN, StarGAN, StyleGAN, StyleGAN2	Mean over unseen GAN architectures: 0.997 AUC, 95.6% accuracy	Yes – seven held-out GAN architectures	Adversarial robustness deferred to future work
Gragnaniello et al. (2021)	Critically evaluates leading GAN-image detectors on compressed, resized and unseen-generator images	Xception, SRNet, Spec, M-Gb, Co-Net, Wang2020 ResNet50, PatchForensics (seven detectors) plus the authors' no-down variant	Train: 362K LSUN real + 362K ProGAN fake; test: ImageNet, COCO, RAISE reals and ~39K synthetic	Train: ProGAN; test: BigGAN, CycleGAN, GauGAN, RelGAN, StarGAN, StyleGAN, StyleGAN2	On uncompressed unseen GANs several methods pass 0.9 AUC but drop sharply after compression or resizing; best no-down variant tops 90% accuracy	Yes – trained on ProGAN; 7 unseen GANs	Authors note "we are still very far from having reliable tools for GAN image detection"; the robustness study is preliminary
Coccomini et al. (2022)	ViT compared with a CNN classifier across forgery types	EfficientNetV2-M vs ViT-Base	ForgeryNet: 15 manipulation methods	GAN, reenactment and swap methods, e.g. ATVG-net, Deepfakes, FaceShifter, FS-GAN, StyleGAN2, Talking Head Video	Trained on three face-transfer methods: EfficientNetV2-M 81.1% on one but 47.0% and 57.0% on the other two; ViT-Base ~62.0% on all three	Yes – subset training, testing on 15 methods	Accuracy on new, unseen methods stays fairly low

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Jeong et al. (2022a)	Bilateral high-pass filtering (frequency and pixel level) to amplify frequency artefacts	BiHPF front-end feeding a ResNet-50 classifier	Real: LSUN, FFHQ; ProGAN images used in training	BigGAN, CycleGAN, ProGAN, StarGAN, StyleGAN, StyleGAN2	Trained on 4-class ProGAN: mean accuracy 84.2% and mean AP 81.8% across unseen GAN models	Yes – ProGAN-trained, tested on 5 unseen GANs	None stated by the authors; only GAN generators are evaluated
Jeong et al. (2022b)	FrePGAN: suppresses frequency artefacts using frequency-level perturbation maps	ResNet-50 deepfake classifier plus a perturbation generator	Train: ProGAN-generated images; test: FFHQ, LSUN, ImageNet, CelebA, COCO and Deepfake sets	BigGAN, CycleGAN, GauGAN, ProGAN, StarGAN, StyleGAN, StyleGAN2	Cross-model, 4-class training: 79.4% average accuracy and 87.2% average precision	Yes – seen and unseen GAN models	Trade-off: frequency artefacts help detect known GANs but weaken cross-GAN generalisation
Ju et al. (2022)	Attention fuses global spatial features with local patch features	Two-branch ImageNet-pretrained ResNet-50 with a patch selection module and multi-head attention	Diverse self-built test set: 19 generation models, 128,424 images	19 models, e.g. BigGAN, Celeb-DF, CycleGAN, FF++, GauGAN, Pix2Pix, ProGAN, StarGAN, StyleGAN/2/3, VAEGAN, WGAN	Unseen models: 91.73% mean AP; mixed setting: 96.91% global AP	Yes – centred on unseen models	Artefacts are harder to spot after high-resolution images are resized
Mandelli et al. (2022)	Orthogonally trained CNNs, ensembled to favour real-image scores	Five EfficientNet-B4 CNNs, ensembled	Train: ~116K real (FFHQ, Metfaces, AFHQ2) + ~200K synthetic; test: StyleGAN3 pairs	FaceVid2Vid, score-based models, StarGAN-v2, StyleGAN2, Taming Transformers; unseen: StyleGAN3	0.9995 AUC over the global test set, unseen StyleGAN3 included	Yes – StyleGAN3 held out of training	The CNNs' orthogonality is checked only empirically, without a theoretical basis
Aghasanli et al. (2023)	Interpretable detection using ViT-feature SVM support vectors as prototypes	Fine-tuned ViT and SVM; repository adds xDNN, KNN, Naive Bayes	Custom sets drawn from ImageNet, FFHQ, Oxford-IIIT Pet	Stable Diffusion and various other diffusion methods	Within-dataset best (fine-tuned ViT + MLP head): Oxford-IIIT Pet 99.5%, image-to-image ImageNet 96.7%, class-conditional ImageNet 99.7%, FFHQ 99.4%; 93.0–97.2% with an SVM on fine-tuned features; 81.3% and 84.0% in cross-dataset tests	No – cross-dataset only; no generator held out	Open datasets are scarce; evaluation on self-built custom datasets only

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Corvi et al. (2023a)	Analysis-only forensic study of spectral/frequency artefacts in synthetic images; no detector proposed	None; power-spectrum, autocorrelation and radial/angular spectral analyses	Real: ImageNet, FFHQ, LAION; fakes from 15 models	GANs: BigGAN, GALIP, StyleGAN-T, StyleGAN2/3; VQ models; DMs: ADM, DALL-E 2, eDiff-I, GLIDE, Score-SDE; latent DMs: LDM, SD, DiT	None; artefacts characterised through spectral analyses, no detection metrics	No – no detector was trained	Post-processing, the authors note, "may significantly modify and hide" these artefacts
Corvi et al. (2023b)	Analyses forensic traces (spectral fingerprints) of diffusion models and evaluates existing detectors	Spec, PatchForensics, Wang2020 (augmentation-trained ResNet-50), Grag2021 (first-layer downsampling removed)	Real: COCO, ImageNet, UCID; fakes from COCO prompts	ADM, DALL-E 2, GLIDE, Latent Diffusion, Stable Diffusion; references: BigGAN, EG3D, ProGAN, StyleGAN2/3	Averaged over 12 test generators (the training GAN and 5 diffusion models included): AUC ~75–93% and accuracy ~52–76% uncompressed; accuracy ~50–61% after resizing and compression	Yes – GAN-trained detectors run on diffusion images	Generalisation is still the chief obstacle; GAN-trained detectors do badly on diffusion images, more so once compressed or resized
Ma et al. (2023)	SeDID: per-step error between deterministic reverse steps and denoising diffusion steps	Statistical variant SeDID_Stat and neural-network variant SeDID_NNs	CIFAR-10, TinyImageNet and CelebA, each at 32x32	Only DDPM	SeDID_NNs, mean over the three datasets: ACC 0.9399, AUC 0.9634	No – DDPM alone; other families absent	Optimal timestep differs between datasets; odd trends on CelebA need more investigation
Ojha et al. (2023)	Real/fake classification by a nearest-neighbour or linear probe on frozen vision-language features; backbone not trained	NN or linear probe on final-layer CLIP ViT-L/14 image features	Train: 720k ProGAN real/fake images; test: ForenSynths, diffusion and autoregressive sets	BigGAN, CRN, CycleGAN, DALL-E, DeepFakes, GauGAN, Glide, Guided, IMLE, LDM, ProGAN, SAN, SITD, StarGAN, StyleGAN	+15.07 mAP and +25.90% accuracy over prior best methods on diffusion and autoregressive models unseen in training	Yes – GAN-trained, evaluated on diffusion/AR	Why frozen CLIP features tell real from fake images apart "remains an open question"
Rahman et al. (2023)	Large-scale dataset; multi-class classifier with reduced filter stride	ConvNeXt-Large (altered stem)	ArtiFact, 2,496,738 images in total: 964,989 real and 1,531,749 fake	25 generation methods: 13 GANs, 7 diffusion models, 5 miscellaneous	Tests 1–3 of IEEE VIP Cup 2022: accuracy 96.04%, 83.00% and 90.60% respectively	Yes – challenge tests use unseen generators	Leading detectors still falter on generators unseen in training

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Sha et al. (2023)	Detects and attributes text-to-image fakes with image-only and hybrid (image plus prompt) classifiers	Image-only: ResNet18; hybrid: CLIP image and text encoders plus a 2-layer MLP	Real: MSCOCO, Flickr30k; fakes by four text-to-image models	DALL-E 2, GLIDE, Latent Diffusion, Stable Diffusion	SD-trained hybrid detector (MSCOCO): accuracy 0.932 on Latent Diffusion, 0.899 on GLIDE, 0.885 on DALL-E 2	Yes – SD-trained, evaluated on three unseen T2I models	Image-only detection on other models (e.g. GLIDE, DALL-E 2) is "far from the design goal"
Tan et al. (2023)	LGrad (Learning on Gradients): pretrained-CNN gradients as a generalised representation of artefacts	Pretrained-CNN transformation model mapping images to gradients, then a classifier	ForenSynths / CNNDetection training data: 20 ProGAN categories, each 18,000 fake and 18,000 real; 1-, 2- or 4-class training	8 test models: BigGAN, CycleGAN, Deepfake, GauGAN, ProGAN, StarGAN, StyleGAN, StyleGAN2	1-class setting, mean across the 8 test models: 86.3% ACC, 92.7% AP; vs FrePGAN: +11.4% ACC, +13.4% AP	Yes – across models, data and unseen domains	Authors' own framing: reliance shifts from the data to the transformation model
Wang et al. (2023a)	DIRE (diffusion reconstruction error): residual of an image versus its DDIM reconstruction	ResNet-50 classifying DIRE maps; reconstruction by ADM (LSUN-Bedroom), 20 DDIM steps	DiffusionForensics benchmark built on LSUN-Bedroom and ImageNet	Diffusion models (8): ADM, DDPM, iDDPM, LDM, PNDM, SD-v1, SD-v2, VQ-Diffusion	ADM-trained detector: 99.9% mean accuracy, 100 AP over 8 diffusion models, unseen ones included	Yes – holds on unseen diffusion models	None stated by the authors; covers only diffusion-generated images
Zhu et al. (2023)	Million-scale benchmark with tasks for cross-generator and degraded-image testing	ResNet-50, DeiT-S, Swin-T, CNNSpot, Spec, F3Net, GramNet	GenImage: >1M fake pairs and 1,331,167 real ImageNet images	ADM, BigGAN, GLIDE, Midjourney, SD v1.4, SD v1.5, VQDM, Wukong	ResNet-50 accuracy: 99.9% on its training generator (SD v1.4), 54.9% on Midjourney	Yes – includes a cross-generator classification task	Steep performance drops across generators and on low-resolution, JPEG-compressed or blurred images
Bird and Lotfi (2024)	Binary CNN real/fake classifier, explained via Grad-CAM	Best of 36 custom CNNs: 2 conv layers of 32 filters, 64-unit dense layer	120,000-image CIFAKE dataset: 60k real (CIFAR-10), 60k synthetic	Stable Diffusion v1.4, a latent diffusion model	Test accuracy of 92.98% on CIFAKE	No – SD v1.4 is the sole generator	One generator at 32px; decisions hinge on background imperfections
Cazenavette et al. (2024)	Stable Diffusion inversion as detection features	From-scratch ResNet50 on the image concatenated with its decoded noise map and reconstruction	Real images: LAION, WebLI, reverse-image-search sets; SynRIS benchmark (new; 15 test sets)	Detector trained on SD; 15+ test models, e.g. DALL-E 2/3, Imagen, Kandinsky 2/3, Midjourney, PixArt-alpha, Playground 2.5, SDXL, Stable Cascade	AUC-ROC 0.759 averaged over the tested generators (RIS protocol, SD+LAION training)	Yes – SD-trained; T2I models unseen in training	Might not generalise beyond diffusion to text-to-image GANs or transformers; inversion costs significantly more compute

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Chen et al. (2024)	DRCT framework: contrastive learning on diffusion-rebuilt hard samples	Backbones: ConvNeXt-B, CLIP-ViT-L/14 (UnivFD)	DRCT-2M (the authors' million-scale set) and GenImage; MSCOCO reals	16 diffusion model types, SD v1.4 and v2 included	Accuracy rises by over 10% in cross-set tests (training on SD v1.4, testing on GenImage subsets)	Yes – cross-set tests on held-out diffusion models	Smaller gains on non-diffusion (GAN) images; covers fully generated images only, not locally generated regions
Cozzolino et al. (2024a)	Few-shot forensics: linear SVM over frozen CLIP features of a few reference images	Linear SVM on penultimate-layer CLIP ViT-L/14 features	Few-shot reference set: 10 to 10k paired real/fake images from one generator; 18-generator test suite	18 in total: ADM, DALL-E 2/3, DeepFloyd-IF, DiT, eDiff-I, Firefly, GigaGAN, GLIDE, LDM, Midjourney v5, ProGAN, Score-SDE, SD variants, StyleGAN2/3/T	~90.0% mean AUC on the 18 generators; over the state of the art, +6% AUC out-of-distribution, +13% on impaired images	Yes – 18 test generators spanning 3 families	Limited interpretability: it remains unclear which forensic cues CLIP relies on
Cozzolino et al. (2024b)	Zero-shot, entropy-based: scores an image's coding cost with a model learnt only from real images	ZED: multi-resolution, lossless neural image encoder	Encoder trained on ~338k real images (Open Images); 19-generator synthetic test set	19, among them ADM, BigGAN, DALL-E 1/2/3, DiT, Firefly, GauGAN, GigaGAN, GLIDE, Midjourney v5, SD/SDXL, StyleGAN2	~90.0% mean AUC on the 19 generators; average accuracy >3% above the state of the art	Yes – 19 never-seen generators, zero-shot	Unreliable under heavy compression or resizing; catches only fully synthetic images, not local edits
Grommelt et al. (2024)	Shows detection datasets carry JPEG-compression and image-size biases, and removes them	Detectors: ResNet50, Swin-T	GenImage	ADM, BigGAN, GLIDE, Midjourney, SD v1.4, SD v1.5, VQDM, Wukong	Bias-corrected training lifts GenImage cross-generator accuracy by >11 pp: ResNet50 from 71.68% to 82.74%, Swin-T from 74.09% to 85.83%	Yes – across GenImage generator subsets	Constraints notwithstanding, "datasets may contain other undesirable biases" such as style differences or generator fingerprints
Khan and Dang-Nguyen (2024)	Universal detection by prompt-tuning CLIP, a vision-language model	Prompt-tuned CLIP ViT versus linear probe, fine-tuning and adapter	Training: 200k ProGAN; testing: 21 datasets	GANs, diffusion models and commercial tools	Prompt tuning: mAP 98.06%, mean accuracy 89.45% on 18 unseen generators	Yes – 21 test datasets, all unseen	Weaker on images from commercial tools once JPEG-compressed
Koutlis and Papadopoulos (2024)	Uses intermediate CLIP encoder-block outputs, weighted by trainable importance scores	RINE: small trainable projection/weighting network over intermediate blocks of a frozen CLIP ViT-L/14	1/2/4-class ProGAN training settings; 20 datasets for testing	BigGAN, CRN, CycleGAN, DALL-E, DeepFake, GauGAN, Glide (3), Guided, IMLE, LDM (3), ProGAN, SAN, SITD, StarGAN, StyleGAN/2	4-class training, 20 test sets: mean accuracy 91.5%, AP 98.8; +10.6% accuracy versus the state of the art	Yes – GANs and diffusion in 20 test sets	Performance drops considerably when blur, crop, compression and noise are applied together

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Liu et al. (2024a)	Forgery-aware CLIP adaptation: image- and frequency-domain adapter with language-guided alignment	FatFormer: CLIP-ViT with forgery-aware adapter and text-prompt alignment	Training: 4-class ProGAN; testing: unseen GAN and diffusion sets	Unseen test-time GANs and diffusion models	Unseen GANs: 98% accuracy; unseen diffusion models: 95%	Yes – held-out GANs and diffusion models	ProGAN-only training; authors note room to do better on diffusion models like Guided and call the GAN-diffusion artefact link unresolved
Luo et al. (2024)	Diffusion-image detection with a LaRE (latent reconstruction error) feature	LaRE feature with EGRE (Error-Guided feature REfinement module; spatial/channel align-then-refine)	GenImage benchmark	GenImage's 8 generator subsets	Surpasses prior best methods by up to 11.9% average ACC and 12.1% average AP over GenImage's 8 generators	Yes – evaluated on 8 GenImage generators	Real/fake features still overlap for generators unseen in training (Midjourney, ADM, BigGAN); LaRE cannot stand alone as input
Ricker et al. (2024a)	Training-free detection from an LDM autoencoder's reconstruction error	Pretrained LDM autoencoders; LPIPS reconstruction distance; no classifier trained	LAION-5B reals with prompt-matched fakes; Midjourney images from Discord	Stable Diffusion 1.1/1.5/2.1, Kandinsky 2.1, Midjourney 4/5/5.1	Mean AP 0.992 over leading LDMs such as SD, Kandinsky, Midjourney	Yes – several LDMs, proprietary Midjourney included	Best results need the source LDM's own autoencoder; limited to latent diffusion
Ricker et al. (2024b)	Benchmarks GAN-image detectors on diffusion images, analyses frequency-domain artefacts, retrains them	Wang2020 (ResNet-50); Gragnaniello2021; Mandelli2022 (an EfficientNet-B4 ensemble)	Mainly LSUN Bedroom 256; also LSUN Cat/Horse, ImageNet, FFHQ	5 GANs: Diff-ProjectedGAN, Diff-StyleGAN2, ProGAN, ProjectedGAN, StyleGAN; 5 DMs: ADM, DDPM, IDDPM, LDM, PNDM	Average AUROC loss of 15.2% for GAN detectors on DM images; Pd@1%FAR averages 94.26% for a DM-trained detector on GANs but only 26.34% for a GAN-trained one on DMs	Yes – GAN-trained and DM-trained detectors cross-tested	DMs leave fewer artefacts to detect; perturbations hurt DM-image detection more
Tan et al. (2024a)	Learning in frequency space: FFT-domain convolution on high-frequency image/feature representations	FreqNet, a lightweight residual-block CNN (1.9M parameters, no pretraining)	Train: ForenSynths ProGAN (1/2/4-class); test: GAN self-synthesis sets	17 GANs, among them AttGAN, BigGAN, CycleGAN, GauGAN, ProGAN, StarGAN, StyleGAN and StyleGAN2; also a Deepfake set	Mean accuracy 92.8% on 17 GAN test sets, beating Ojha (83.0%) by +9.8%	Yes – ProGAN-trained, evaluated on 17 GAN sources	The conclusion names no limitations; only GAN sources are evaluated

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Tan et al. (2024b)	Up-sampling artefacts captured via Neighbouring Pixel Relationships (NPR)	Lightweight CNN on NPR input; conv layers and ResNet blocks, 1.44M parameters	ForenSynths ProGAN 4-class training split (car, cat, chair, horse); 18K real/18K fake each	28 generators: 16 GANs, 11 diffusion models, 1 deepfake method	Averages 92.2% accuracy on the 28 generators, +11.6% versus the previous best method	Yes – 28 held-out GAN and diffusion models	None stated explicitly; training used only ProGAN 4-class data
Tsigos et al. (2024)	Quantitative framework for scoring how XAI methods explain a deepfake detector	Grad-CAM++, RISE, SHAP, LIME, SOBOL benchmarked on an EfficientNetV2 detector	FaceForensics++	Four face manipulation methods: DeepFakes, Face2Face, FaceSwap, NeuralTextures	LIME best: detector accuracy left at 0.579 on NeuralTextures after altering its top segment, 0.245 on DeepFakes after its top three	No – one detector, FF++ alone	Explanation methods still struggle most with Face2Face and FaceSwap fakes
Park and Owens (2025)	Supervised detector learnt from an extremely varied many-generator dataset	CLIP-pretrained ViT-S classifier; ConvNeXt-S also trialled	Training: Community Forensics (4,803 generators, 2.7M images); testing: Wang, Ojha, Synthbuster, GenImage, authors' 21-model set	4,803 in total: latent/pixel diffusion, GANs, commercial systems	On the out-of-distribution set of 21 unseen models: 0.987 mAP, 89.2% accuracy	Yes – 21 held-out models from 5 families	Training data leans towards diffusion models; errors remain too frequent for critical uses
Tan et al. (2025)	Injects a category-common prompt into CLIP's text encoder so that image features carry detection semantics	C2P-CLIP: CLIP ViT-L/14 plus a category common prompt	UniversalFakeDetect (train on ProGAN; 19 test subsets); GenImage (train on SD v1.4)	20 in total: UniversalFakeDetect subsets plus GenImage diffusion models	UniversalFakeDetect (19 test subsets): accuracy 93.79%, mAP 98.66%; GenImage: 95.8% accuracy	Yes – spans 20 generation models	Prompt analysis by word frequency alone, without full captions, gives incomplete information
Yan et al. (2025a)	Mixture-of-experts fusion of frequency/noise (low-level) and semantic (high-level) cues; proposes Chameleon, an in-the-wild benchmark	AIDE: SRM noise filters, a DCT scoring module and an OpenCLIP semantic encoder	AIGCDetectBenchmark, GenImage, Chameleon (authors' own)	ADM, BigGAN, CycleGAN, DALLE-2/3, GauGAN, Glide, Midjourney, ProGAN, SD v1.4/1.5, StarGAN, StyleGAN/2, VQDM, Wukong	Mean accuracy 92.77% (AIGCDetectBenchmark) and 86.88% (GenImage), but only 65.77% on Chameleon	Yes – 16+ generators, several benchmarks	Every tested detector nearly fails on Chameleon's realistic images; the problem remains "far from being solved"

Study	Detection approach	Model / architecture	Dataset used	Generator(s) covered	Reported accuracy / AUC	Cross-generator tested?	Key limitation
Yan et al. (2025b)	Effort: SVD splits frozen foundation-model features into orthogonal subspaces	CLIP ViT-L/14 (ablations: BEiT-v2, SigLIP)	Training: ProGAN; testing: 19 UniversalFakeDetect subsets, plus FF++ (deepfakes)	BigGAN, CycleGAN, DALL-E, GauGAN, GLIDE, Guided/Latent Diffusion, ProGAN, StarGAN, StyleGAN	mAP 99.41%, mAcc 95.19% over 19 test subsets, unseen generators among them	Yes – unseen generators among the 19 test subsets	Authors raise only misuse risk and discuss no technical limitations

Note: Each row is a shortened version of the study's record in the review's extraction table. "Cross-generator tested?" says whether the detector was evaluated on generators that were absent from its training data; "Key limitation" mostly gives the weakness that the study's own authors acknowledge; some cells carry a scope point noted by this review, for instance where the authors state no limitation.

12.1 What the Comparison Table Shows

Table 12.2 gathers the main tallies, which the paragraphs that follow discuss in turn.

Table 12.2. Tallies drawn from Table 12.1

Measure	Count
Studies reporting a cross-generator evaluation	36 of 48 (12 do not)
Generator coverage: GAN-era generators only	20
Generator coverage: diffusion-family generators only	8
Generator coverage: both	19
Generator coverage: face-manipulation methods only	1
Studies trained on ProGAN-generated images (14 of them on the ForenSynths training set)	15
Studies quoting an accuracy figure	roughly 38
Studies quoting average precision	15
Studies quoting AUC or AUROC	9
Studies reporting no detection metric	1
Studies giving an accuracy figure and nothing else	around 23
Studies naming robustness to degraded or low-resolution input as a limitation	10
Studies including an explanation analysis	4

Cross-generator testing has become common, though not universal: it appears in 36 of the 48 studies and is missing from 12. Among those twelve, three are early spectral studies evaluated dataset by dataset (Durall et al., 2019; Durall et al., 2020; Dzanic et al., 2020), two attribute images among a closed set of known GANs (Marra et al., 2019; Yu et al., 2019), and three are limited to one generator or one detector (Ma et al., 2023; Bird and Lotfi, 2024; Tsigos et al., 2024). The other four are an analysis paper that proposes no detector (Corvi et al., 2023a), a counter-study that trains a separate classifier for each GAN (Chandrasegaran et al., 2021), a DCT analysis in which no GAN is held out (Frank et al., 2020), and a prototype-based method whose cross-dataset test uses Stable Diffusion images on both sides (Aghasanli et al., 2023).

To count coverage, each study was classified by the generator families on its list, with benchmark names expanded into the generator lists of Table 10.1. On this count, 20 studies use only GAN-era generators, 8 use only diffusion-family generators, 19 use both, and one study covers nothing but face-manipulation methods (Tsigos et al., 2024). Training data is concentrated: fifteen studies, close to a third of all those

surveyed, train on ProGAN-generated images (Chai et al., 2020; Wang et al., 2020; Cozzolino et al., 2021; Gragnaniello et al., 2021; Jeong et al., 2022a; Jeong et al., 2022b; Ojha et al., 2023; Tan et al., 2023; Khan and Dang-Nguyen, 2024; Koutlis and Papadopoulos, 2024; Liu et al., 2024a; Tan et al., 2024a; Tan et al., 2024b; Tan et al., 2025; Yan et al., 2025b). Fourteen of them use the ProGAN training set released with ForenSynths; the exception, Chai et al. (2020), trains on ProGAN-generated face images instead.

Metrics are concentrated similarly. Accuracy is quoted by roughly 38 studies, average precision by 15 and AUC or AUROC by 9; one study reports no detection metric of any kind (Corvi et al., 2023a). Around 23 quote accuracy alone, a figure that, as Chapter 11 noted, hides a choice of decision threshold. Ten studies' stated limitations concern robustness to degraded or low-resolution images, the degradation being compression, resizing, blurring or other post-processing (Durall et al., 2019; Dzanic et al., 2020; Ju et al., 2022; Zhu et al., 2023; Corvi et al., 2023a; Corvi et al., 2023b; Khan and Dang-Nguyen, 2024; Koutlis and Papadopoulos, 2024; Cozzolino et al., 2024b; Ricker et al., 2024b).

12.2 The Chronological Arc

In order of publication, the table shows a clear progression. The first rows pair a single artefact with a simple classifier and report near-perfect numbers. Co-occurrence matrices give above 99% (Nataraj et al., 2019) and DCT coefficients 100.00% on FFHQ (Frank et al., 2020); a spectral-decay feature reaches 99.2% for uncompressed images at 1024×1024 (Dzanic et al., 2020); and one study scores 100% on a face set its authors assembled, even when only 20 samples were labelled (Durall et al., 2019). Such figures hold only in narrow conditions, and qualifying rows follow quickly. At 256×256 the same feature manages only 88.8%, and under 85% JPEG compression it becomes unusable (Dzanic et al., 2020). Retraining the generators with bilinear upsampling cuts the detection rate of detectors relying on that feature from 84.8–99.8% to 0–0.3% (Chandrasegaran et al., 2021).

Transfer-learning CNNs enter in the same years with a wider claim. A ResNet-50 trained on ProGAN images alone averages 90.8 mAP over eleven test generators, ten of which it never saw (Wang et al., 2020). A contrastively trained detector averages an AUC of 0.997 and an accuracy of 95.6% on seven unseen GANs (Cozzolino et al.,

2021), and an ensemble that kept StyleGAN3 out of training reaches 0.9995 AUC (Mandelli et al., 2022). Yet Gragnaniello et al. (2021), evaluating seven state-of-the-art GAN-image detectors side by side, concluded that reliable tools were still far off.

Diffusion models then wiped out most of this lead, a collapse that two rows record. In the first, four GAN-trained detectors were run on twelve generators, five of them diffusion models (ADM, DALL·E 2, GLIDE, Latent Diffusion and Stable Diffusion); across all twelve they averaged roughly 52–76% accuracy without compression and about 50–61% once resizing and compression were added, and the diffusion models were among the hardest cases (Corvi et al., 2023b). The second reports an average drop of 15.2% in AUROC when such detectors meet diffusion output, and shows that the effect depends on direction: trained on diffusion images, a detector still reaches an average Pd@1%FAR of 94.26% on GAN images, whereas the reverse arrangement gives only 26.34% (Ricker et al., 2024b). The same failure appears even when both generators come from the diffusion family: in the GenImage benchmark, a ResNet-50 trained on Stable Diffusion v1.4 images scores 99.9% on test images from that generator but 54.9% on the Midjourney test set (Zhu et al., 2023).

Several 2023–2024 rows answer by deriving a signal from each image and handing that signal, with or without the image, to the detector. LGrad passes it gradients from a pretrained network and averages 86.3% accuracy across the eight models it was tested on (Tan et al., 2023); NPR uses relationships between neighbouring pixels and reaches 92.2% across 28 generators (Tan et al., 2024b); DIRE reconstructs each image with DDIM and classifies the residual between the two, averaging 99.9% accuracy across eight diffusion models (Wang et al., 2023a). AEROBLADE dispenses with the trained classifier, scoring each image by its distance from an autoencoder reconstruction for a mean AP of 0.992 (Ricker et al., 2024a). LaRE² computes the same kind of error in latent space (Luo et al., 2024), while FakeInversion draws on the inversion of Stable Diffusion, recording an average AUC-ROC of 0.759 on text-to-image models absent from its training data (Cazenavette et al., 2024).

The latest rows converge on a further idea: keep a large general-purpose network frozen and train only a small component on top of it. Ojha et al. (2023) established the idea by classifying frozen CLIP features with a linear probe or a nearest-neighbour rule. On diffusion and autoregressive models absent from training, this raised

accuracy by 25.90% and mAP by 15.07 over the earlier best. Later rows use these features in other ways:

- a few-shot linear SVM, averaging about 90.0% AUC across 18 generators (Cozzolino et al., 2024a);
- intermediate encoder-block outputs, giving 91.5% accuracy with 98.8 AP on 20 test sets (Koutlis and Papadopoulos, 2024);
- a trainable adapter, which scores 98% on GANs held out of training and 95% on held-out diffusion models (Liu et al., 2024a);
- prompt tuning, with 98.06% mAP across 18 generators held out of training (Khan and Dang-Nguyen, 2024);
- a prompt injected into the text encoder, reaching 93.79% accuracy across 19 test subsets (Tan et al., 2025);
- an orthogonal subspace decomposition, with 95.19% mean accuracy and 99.41% mAP on the same 19 subsets (Yan et al., 2025b).

12.3 What the Strongest Cross-Generator Results Share

The rows with the best unseen-generator results tend to share three properties. First, the representation is not learned from detection data. The foundation-model rows freeze the backbone and train only a prompt, an adapter or a probe (Ojha et al., 2023; Khan and Dang-Nguyen, 2024; Koutlis and Papadopoulos, 2024; Cozzolino et al., 2024a; Liu et al., 2024a; Tan et al., 2025; Yan et al., 2025b), and ZED, trained on real images alone, still averages about 90.0% AUC over 19 generators (Cozzolino et al., 2024b). Second, the decision does not rest on the raw pixels alone but on something computed from the image, such as gradient maps (Tan et al., 2023), relations between neighbouring pixels (Tan et al., 2024b) or signals derived from reconstructing it (Wang et al., 2023a; Cazenavette et al., 2024; Luo et al., 2024; Ricker et al., 2024a). Third, the training side is diverse. A detector trained on Community Forensics, which holds 2.7M images produced by 4,803 generators, reaches 89.2% accuracy and 0.987 mAP on a set of 21 models it was not trained on (Park and Owens, 2025); and training data matter in a second way, since once the JPEG-compression and image-size biases in GenImage are removed, the cross-generator accuracy of both a ResNet50 and a Swin-T rises by more than eleven points (Grommelt et al., 2024). In none of the three cases does a network have to learn, from scratch, an artefact that belongs to a single generator.

12.4 What the Headline Figures Hide

These headline figures must be read alongside the two columns to their right. Most record one move from training to test data, often starting from ProGAN-generated images, the training source of fifteen rows, so they measure a single transfer and not transfer in general. Averages also mask the worst case: a classifier built on the frequency spectrum, with a 97.2% average, breaks down completely on GauGAN (Zhang et al., 2019), and the authors of LaRE² concede that real and fake features still overlap for Midjourney, ADM and BigGAN images (Luo et al., 2024). Performance on benchmarks and in the wild can differ sharply. The detector proposed with the Chameleon benchmark averages 92.77% on AIGCDetectBenchmark and 86.88% on GenImage but only 65.77% on Chameleon's realistic images, where every detector its authors tested comes close to failing (Yan et al., 2025a). The ten stated limitations about degraded input are supported by the compression and resizing results given in Section 12.2 (Corvi et al., 2023b) and by a zero-shot detector that its authors describe as unreliable once images are heavily compressed or resized (Cozzolino et al., 2024b). Chapters 10 and 11 traced the dataset and metric causes; here the point is that, in Table 12.1, the rightmost columns qualify the headline one.

12.5 Where Explainability Appears

Explanations are analysed in just four of the 48 rows, usually after the detector has been built. Bird and Lotfi (2024) applied Grad-CAM to a small CNN trained on CIFAKE and found its decisions driven by imperfections in the background. Chai et al. (2020) produced a map of the patches in which a fake can be detected, by limiting how much of the image each decision sees. Aghasanli et al. (2023) instead built interpretability into the detector, using the support vectors of an SVM fitted on fine-tuned ViT features as prototypes. Tsigos et al. (2024) compared five explanation methods on a single detector and rated LIME best, because altering the segments it ranked highest lowered the detector’s accuracy the most (to 0.579 on NeuralTextures after one segment, and to 0.245 on DeepFakes after three). Of these four, only Chai et al. (2020) also tested their detector on unseen generators; the other three carry No in the cross-generator column of Table 12.1. On this evidence, explainability and generalisation are pursued as separate questions.

12.6 The Combination No Study Reports

Architectures are compared under a single protocol in a few rows. The GenImage benchmark tests ResNet-50, DeiT-S and Swin-T classifiers alongside four detectors built for the task (Zhu et al., 2023), and Grommelt et al. (2024) applied one bias-correction procedure to both a ResNet50 and a Swin-T. Coccomini et al. (2022) came nearest to isolating architecture alone: they trained an EfficientNetV2-M and a ViT-Base on identical ForgeryNet subsets and tested both on fifteen manipulation methods. In one of their experiments the CNN reached 81.1% on one of its three training methods but only 47.0% and 57.0% on the other two, while the ViT stayed near 62.0% on all three; overall, the authors found that the CNN specialises more and generalises worse to new methods. Still, none of them compares several of the ImageNet-pretrained CNN backbones discussed in Section 9.2, and none analyses explanations.

The opposite combination is absent as well. The only row built on CIFAKE relies on a custom CNN, the best of 36 topologies, instead of pretrained backbones; it covers a single generator and offers Grad-CAM explanations without any cross-generator result (Bird and Lotfi, 2024). The interpretable, prototype-based row also carries a No (Aghasanli et al., 2023), and the only study that scores explanation quality numerically looks at just one detector, on FaceForensics++ (Tsigos et al., 2024). As a result, no row of Table 12.1 combines all four of these elements:

- several CNN backbones pretrained on ImageNet, all trained under a common protocol;
- a Vision Transformer, trained the same way, as a benchmark;
- an explanation analysis showing which image regions the trained models focus on;
- an evaluation on generators withheld from training.

Each element is well established in at least one row. What no surveyed study reports is all four at once, on a dataset small enough for such a matched comparison to stay manageable. The gap is this combination rather than any one element, and Chapter 13 builds the gap analysis on it.

13. CHALLENGES AND GAP ANALYSIS

Chapters 9 to 12 covered the detectors that have been built and how their performance is measured. This chapter turns to the problems those chapters leave unsolved. It groups them into five named sub-problems and ends with the research gap behind the planned second phase of the project.

13.1 Cross-Generator Generalisation

Chapter 7 argued that weak transfer to new generators is a structural problem and not simply an engineering nuisance. The evidence that a detector learns may be a side effect of something specific to one generator: its upsampling operator, its training run or its decoder. If so, a new generator that alters any of these does more than supply harder examples: it takes away the detector's evidence.

In the GAN era, transfer looked dependable at first: a ResNet-50 trained with ProGAN as its only generator averaged 90.8 mAP across eleven test generators, ten of which it had never seen (Wang et al., 2020). Diffusion models brought this period to an end. Detectors trained on GAN output lose an average of 15.2% AUROC when they are tested on diffusion images (Ricker et al., 2024b), and in another study four GAN-trained detectors averaged only about 52–76% accuracy over twelve generators that included five diffusion models (Corvi et al., 2023b). The same weakness then appeared within the diffusion family: in the GenImage benchmark, a ResNet-50 tested on its training generator is 99.9% accurate, while the same model tested on Midjourney is just 54.9% accurate (Zhu et al., 2023). Lastly, the gap does not only divide generator families; it also separates benchmark data from what a detector would face in practice. On realistic in-the-wild images, a detector with a mean accuracy of 92.77% on a standard benchmark falls to 65.77% (Yan et al., 2025a). Section 12.3 found three features shared by the strongest cross-generator rows: frozen general-purpose representations, derived inputs and diversity in the training data. None of them closes the gap.

13.2 Robustness to Compression and Post-Processing

Robustness is a weak point that authors often acknowledge in Table 12.1: in ten of the 48 studies, the authors' own stated limitation concerns degraded or low-resolution input (Section 12.1). The reason again lies in the nature of the evidence. Generator

traces are concentrated at high frequencies, the very part of the signal that lossy compression removes (Section 7.3). At 85% JPEG compression, the spectrum of a generated image becomes essentially impossible to tell apart from that of a real one (Dzanic et al., 2020); routine post-processing is also able to alter or conceal the artefacts (Corvi et al., 2023a).

The effect has also been measured directly. Four GAN-trained detectors averaged roughly 52–76% accuracy over twelve generators on uncompressed images, and only about 50–61% once the same images had been resized and compressed (Corvi et al., 2023b). A zero-shot detector (Section 9.2) averages about 90.0% AUC, yet its own authors call it unreliable when inputs have been heavily compressed or resized (Cozzolino et al., 2024b). The accuracy of a CLIP-feature detector also falls once blurring, cropping, compression and added noise are applied together (Koutlis and Papadopoulos, 2024). Compression is not a form of attack; it is applied by default on every platform that a suspect image would pass through.

13.3 Dataset Bias and Staleness

A detector can score well on a dataset without detecting anything at all. Grommelt et al. (2024) found two systematic differences between the real and the generated images of GenImage, one in JPEG compression and one in image size. A classifier can therefore score well by picking up these two properties instead of any forensic trace. Once both biases are removed, cross-generator accuracy improves by over eleven points, both for a ResNet50 and for a Swin-T (Grommelt et al., 2024).

Datasets built from one generator raise a narrower form of the same problem. CIFAKE, for instance, covers a single generator at 32×32 resolution. The Grad-CAM analysis that accompanies the dataset finds that the trained network bases its decisions on small imperfections in the image backgrounds, which are a property of that generator and not a general sign of synthesis (Bird and Lotfi, 2024). With only one generator, a dataset of this kind cannot reveal a cross-generator drop in the first place. Staleness has the same root. Each dataset freezes its list of generators when it is published. Apart from a few in-the-wild collections (Section 10.1), the field has answered by making the lists longer rather than by moving away from the fixed-list format. The gap between benchmark and in-the-wild performance that Yan et al. (2025a) report is what this choice eventually costs.

13.4 Interpretability

Of the 48 rows in Table 12.1, only four include any form of explanation (Section 12.5), and these four mostly offer post-hoc saliency maps. Such a map, computed for an already trained network, shows where the network looked but not what it found there. The difference matters more in detection than in object recognition, since the traces a detector relies on are often invisible to the eye.

Applied to the CIFAKE network, Grad-CAM exposes a shortcut, the reliance on background imperfections noted in Section 13.3, but it does not identify which property of the highlighted pixels is synthetic (Bird and Lotfi, 2024). A prototype-based classifier is interpretable by design, yet its cross-dataset test used the same generator on both the training and the test side (Aghasanli et al., 2023). The only row that compares explanation methods quantitatively is confined to one detector and one dataset (Tsigos et al., 2024). Frozen-feature detectors are furthest from an explanation: their own authors note that it is still an open question why a CLIP representation tells real images apart from generated ones (Ojha et al., 2023; Cozzolino et al., 2024a). Revealing a shortcut has value, yet it does not amount to stating what a detector measures.

13.5 Reproducibility and Comparability

One further, smaller problem makes the others worse. Chapter 11 noted that studies report different combinations of metrics on different test suites, which means two headline figures seldom measure the same quantity. Around 23 of the 48 studies give an accuracy figure and no other metric (Section 12.1). Cross-study comparisons are therefore weaker evidence than they first appear. This is a reason to compare candidate architectures under one common protocol rather than to piece together figures from separate papers.

13.6 The Research Gap

Section 12.6 identified the combination that is missing. No row of Table 12.1 brings together all four of the following elements:

1. Several ImageNet-pretrained CNN backbones, all trained with one protocol.
2. A Vision Transformer benchmark trained with that same protocol.
3. An analysis that shows which image regions the trained models pay attention to.

4. A cross-generator test, on generators the models did not see in training.

Each element can be found in at least one row. The GenImage benchmark places a ResNet-50 beside transformer backbones (Zhu et al., 2023); Coccomini et al. (2022) change only the architecture and hold the training data constant; and the CIFAKE study includes a Grad-CAM analysis (Bird and Lotfi, 2024). No row, however, holds all four. The study planned to address this gap is set out in Section 14.3.

14. CONCLUSION, LIMITATIONS AND FUTURE SCOPE

This chapter brings the report to a close. Section 14.1 sums up what the review found, Section 14.2 sets out the limits within which those findings hold, and Section 14.3 describes the future scope of the work, beginning with the study planned for the Semester 8 major project.

14.1 Conclusion

This review examined the deep-learning methods that have been proposed for detecting AI-generated still images. It organised the field into five families according to where a detector's evidence comes from (Chapter 9): handcrafted, statistical and frequency-domain forensics; convolutional networks adapted through transfer learning; transformer-based and attention-based models; detectors that reuse frozen vision-language representations; and work that sets out to explain detection decisions. The review went on to compare the datasets and benchmarks most often used for evaluation (Chapter 10) and the metrics in which results are reported (Chapter 11). Its central comparison placed all 48 primary studies in one master table (Table 12.1). Besides the usual entries, that table carries two further columns, *Cross-generator tested?* and *Key limitation*: the first shows whether a detector was tested on generators missing from its training data, and the second gives the limitation that the study's own authors acknowledge.

The comparison leads to three main findings. First, the generalisation gap is structural and not incidental. A large part of what a detector picks up during training is a by-product of the design of one particular generator. When that design changes, the evidence disappears instead of merely becoming harder to find. The drops reported in the literature are steep: one detector's accuracy falls from 99.9% on its training generator to 54.9% when its test images come from Midjourney instead (Zhu et al., 2023), and a mean accuracy of 92.77% on a standard benchmark becomes 65.77% on realistic in-the-wild images (Yan et al., 2025a). Second, the best cross-generator results are linked by three recurring properties: the representation was not trained on detection data, the detector receives an input computed from the image instead of the image itself, or the training data span many generators. Third, testing on unseen generators and

analysing explanations seldom meet in one study. Of the 48 primary studies, 36 include a cross-generator evaluation, but only four include an explanation analysis.

This third finding forms one part of the research gap stated in Section 13.6. The experimental study designed to fill that gap is described in Section 14.3.

14.2 Limitations of This Review

The conclusions in Section 14.1 hold only within limits that come from the design of the review, and they should be read with these limits in mind.

The first set of limits concerns coverage. The literature search was performed once, in Week 2 (27 July to 2 August 2026), so any study published after that week lies outside the surveyed set. Under the protocol, only work written in English and available in full text could be included, and detection studies had to date from 2019 to 2026; background references on generators, backbones and explanation methods were exempt from that window. The included studies were also not all read to the same depth. Full-text extraction covered the 48 primary studies, whereas the 11 secondary studies were consulted selectively and have no row in Table 12.1.

The second set concerns the evidence. All performance figures in this report come from the original publications; each was obtained under its own authors' protocol, and none was reproduced for this review. Because the studies combine different metrics with different test suites, their results can be compared directly only with caution (Section 11.2). Lastly, no inter-rater agreement statistic was calculated for screening or for extraction. A statistic of this kind measures how closely independent reviewers agree in their decisions; without one, the consistency of these judgements remains unquantified.

14.3 Future Scope

The Semester 8 major project is planned around the combination that Section 13.6 found missing. At its centre is a comparison, based on transfer learning, of four ImageNet-pretrained CNN backbones, namely VGG19, ResNet50, DenseNet121 and EfficientNet-B0 (Simonyan and Zisserman, 2015; He et al., 2016; Huang et al., 2017; Tan and Le, 2019). Every backbone is to be used in two configurations. In feature extraction, the pretrained layers stay frozen and only a classifier placed on top of them is trained; in fine-tuning, the pretrained weights are updated as well. A Vision

Transformer (Dosovitskiy et al., 2021), trained under the same conditions as the CNNs, will act as the benchmark for the comparison.

Each element of this design follows from a finding of the review, and Table 14.1 lists the elements next to the findings behind them. The four backbones make up the family that Section 9.2 identified as the standard choice for transfer learning, although its members are rarely compared with one another on equal terms (Section 12.6). Holding every model to one matched protocol responds to Section 13.5, which argued that architectures should be compared under a common protocol and not through figures gathered from separate papers. The ViT is included as a benchmark and not as a proposed improvement, because Section 9.3 found that when a published study changes its backbone, it usually changes the pretraining data and the training objective at once. Explanations will come from Grad-CAM (Selvaraju et al., 2017), applied to incorrect as well as correct predictions, since Section 9.5 showed that an explanation can expose a shortcut while still failing to name the evidence itself. The primary dataset is to be CIFAKE (Bird and Lotfi, 2024). Its 120,000 images keep it small enough for a matched comparison of five architectures to remain feasible on modest hardware, but all of its generated images come from a single generator. A subset of GenImage (Zhu et al., 2023) will therefore supply the cross-generator test, built from generators that the models never saw in training. It is this test that would show whether a result describes one generator or detection more generally.

Table 14.1. Design of the planned Semester 8 study and the review finding behind each element

Element	Planned choice	Review finding that motivates it
CNN backbones	ImageNet-pretrained ResNet50, DenseNet121, EfficientNet-B0 and VGG19	The standard transfer-learning family, rarely compared on equal terms (Sections 9.2 and 12.6)
Training configurations	Feature extraction and fine-tuning for each backbone, under one matched protocol	Figures from separate papers seldom measure the same thing (Section 13.5)
Transformer benchmark	Vision Transformer trained under the same conditions, as a reference point	A backbone change in published work usually also changes pretraining data and objective (Section 9.3)
Explanation method	Grad-CAM on correct and incorrect predictions	An explanation can expose a shortcut without naming the evidence (Section 9.5)
Primary dataset	CIFAKE (120,000 images)	Limited to one generator (Section 13.3), but small enough for a matched comparison on modest hardware (Section 10.1)

Element	Planned choice	Review finding that motivates it
Cross-generator test	GenImage subset with generators not seen in training	A single-generator dataset cannot reveal a cross-generator drop (Sections 13.1 and 13.3)

On the guide's instruction of 15 August 2026, preparation of the code for this study began as a separate work track, which belongs to the Semester 8 major project and will be reported there.

Beyond this study, the challenges set out in Chapter 13 suggest three wider directions for the field. Detectors should be tested routinely on compressed and post-processed images, since the platforms through which suspect images travel compress them by default (Section 13.2). Evaluation should also move past datasets whose generator lists are frozen at publication towards realistic in-the-wild images, on which detectors that do well on standard benchmarks fall sharply (Sections 13.1 and 13.3), and the quality of explanations should be measured quantitatively, which, among the surveyed studies, has been attempted for only one detector on one dataset (Section 13.4). These directions, like the planned study, lie beyond the present review, which contains no experimental result of the team's own.

15. INDIVIDUAL CONTRIBUTION

This chapter states how the work of Group 148 was shared among its three members, Rohit Yadav, Vishal and Hardik Mehlawat, all students of B.Tech Computer Science and Engineering in the seventh semester. The project was carried out jointly. By the team's own decision, the review was not divided into personal assignments, and no member was made solely responsible for any part of it.

Instead, each of the three took part in every work package listed in Table 6.2: searching for and screening the literature, reading the primary studies in full and extracting their data, building the taxonomy and carrying out the comparative and gap analysis, writing the text together with its tables and figures, verifying the references and checking the records for consistency, and preparing the project documentation, from the weekly progress reports to this final report. The weekly progress reports were likewise submitted for the group as a whole, each listing all three members. The main decisions about the review were taken by the team together, under the guide's direction, rather than by one member acting alone.

Because each member's contribution covers the whole project, Table 15.1 gives the same joint entry against every name, alongside the member's enrollment number.

Table 15.1. Individual contribution

Name	Enrollment No.	Contribution
Rohit Yadav	A2305223472	Joint contribution to all work packages in Table 6.2
Vishal	A2305223568	Joint contribution to all work packages in Table 6.2
Hardik Mehlawat	A2305223449	Joint contribution to all work packages in Table 6.2

REFERENCES

Aghasanli, A., Kangin, D. and Angelov, P. (2023). Interpretable-Through-Prototypes Deepfake Detection for Diffusion Models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)* (pp. 467–474). https://doi.org/10.1109/ICCVW60793.2023.00053

Bammey, Q. (2024). Synthbuster: Towards Detection of Diffusion Model Generated Images. *IEEE Open Journal of Signal Processing, 5*, 1–9. https://doi.org/10.1109/OJSP.2023.3337714

Baraldi, L., Cocchi, F., Cornia, M., Baraldi, L., Nicolosi, A. and Cucchiara, R. (2024). Contrasting Deepfakes Diffusion via Contrastive Learning and Global-Local Similarities. In *Proceedings of the European Conference on Computer Vision (ECCV)*. https://arxiv.org/abs/2407.20337

Bird, J. J. and Lotfi, A. (2024). CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images. *IEEE Access, 12*, 15642–15650. https://doi.org/10.1109/ACCESS.2024.3356122

Cazenavette, G., Sud, A., Leung, T. and Usman, B. (2024). FakeInversion: Learning to Detect Images from Unseen Text-to-Image Models by Inverting Stable Diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2406.08603

Chai, L., Bau, D., Lim, S.-N. and Isola, P. (2020). What makes fake images detectable? Understanding properties that generalize. In *Proceedings of the European Conference on Computer Vision (ECCV)*. https://arxiv.org/abs/2008.10588

Chandrasegaran, K., Tran, N.-T. and Cheung, N.-M. (2021). A Closer Look at Fourier Spectrum Discrepancies for CNN-generated Images Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2103.17195

Chang, Y.-M., Yeh, C., Chiu, W.-C. and Yu, N. (2023). AntifakePrompt: Prompt-Tuned Vision-Language Models are Fake Image Detectors. *arXiv preprint* arXiv:2310.17419. https://arxiv.org/abs/2310.17419

Chattopadhyay, A., Sarkar, A., Howlader, P. and Balasubramanian, V. N. (2018). Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)* (pp. 839–847). https://doi.org/10.1109/WACV.2018.00097

Chen, Y. and Zou, J. (2023). TWIGMA: A dataset of AI-Generated Images with Metadata From Twitter. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. https://arxiv.org/abs/2306.08310

Chen, B., Zeng, J., Yang, J. and Yang, R. (2024). DRCT: Diffusion Reconstruction Contrastive Training towards Universal Detection of Diffusion Generated Images. In *Proceedings of the 41st International Conference on Machine Learning (ICML)* (Vol. 235, pp. 7621–7639). PMLR. https://proceedings.mlr.press/v235/chen24ay.html

Coccomini, D. A., Caldelli, R., Falchi, F., Gennaro, C. and Amato, G. (2022). Cross-Forgery Analysis of Vision Transformers and CNNs for Deepfake Image Detection. In *Proceedings of the 1st International Workshop on Multimedia AI against Disinformation (MAD)* (pp. 52–58). https://doi.org/10.1145/3512732.3533582

Coccomini, D. A., Esuli, A., Falchi, F., Gennaro, C. and Amato, G. (2024). Detecting images generated by diffusers. *PeerJ Computer Science, 10*, e2127. https://doi.org/10.7717/peerj-cs.2127

Corvi, R., Cozzolino, D., Poggi, G., Nagano, K. and Verdoliva, L. (2023a). Intriguing properties of synthetic images: from generative adversarial networks to diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. https://arxiv.org/abs/2304.06408

Corvi, R., Cozzolino, D., Zingarini, G., Poggi, G., Nagano, K. and Verdoliva, L. (2023b). On the detection of synthetic images generated by diffusion models. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. https://arxiv.org/abs/2211.00680

Cozzolino, D., Gragnaniello, D., Poggi, G. and Verdoliva, L. (2021). Towards Universal GAN Image Detection. In *Proceedings of the IEEE International Conference on Visual Communications and Image Processing (VCIP)*. https://arxiv.org/abs/2112.12606

Cozzolino, D., Poggi, G., Corvi, R., Nießner, M. and Verdoliva, L. (2024a). Raising the Bar of AI-generated Image Detection with CLIP. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. https://arxiv.org/abs/2312.00195

Cozzolino, D., Poggi, G., Nießner, M. and Verdoliva, L. (2024b). Zero-Shot Detection of AI-Generated Images. In *Proceedings of the European Conference on Computer Vision (ECCV)*. https://arxiv.org/abs/2409.15875

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J. and Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *Proceedings of the International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2010.11929

Durall, R., Keuper, M., Pfreundt, F.-J. and Keuper, J. (2019). Unmasking DeepFakes with simple Features. *arXiv preprint* arXiv:1911.00686. https://arxiv.org/abs/1911.00686

Durall, R., Keuper, M. and Keuper, J. (2020). Watch your Up-Convolution: CNN Based Generative Deep Neural Networks are Failing to Reproduce Spectral Distributions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2003.01826

Dzanic, T., Shah, K. and Witherden, F. (2020). Fourier Spectrum Discrepancies in Deep Network Generated Images. In *Advances in Neural Information Processing Systems (NeurIPS)* (Vol. 33, pp. 3022–3032). https://arxiv.org/abs/1911.06465

Frank, J., Eisenhofer, T., Schönherr, L., Fischer, A., Kolossa, D. and Holz, T. (2020). Leveraging Frequency Analysis for Deep Fake Image Recognition. In *Proceedings of the International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2003.08685

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A. and Bengio, Y. (2014). Generative Adversarial Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/1406.2661

Gragnaniello, D., Cozzolino, D., Marra, F., Poggi, G. and Verdoliva, L. (2021). Are GAN generated images easy to detect? A critical analysis of the state-of-the-art. In *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*. https://arxiv.org/abs/2104.02617

Grommelt, P., Weiss, L., Pfreundt, F.-J. and Keuper, J. (2024). Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets. In *Proceedings of the European Conference on Computer Vision Workshops (ECCVW)*. https://arxiv.org/abs/2403.17608

He, K., Zhang, X., Ren, S. and Sun, J. (2016). Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/1512.03385

Ho, J., Jain, A. and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2006.11239

Hong, Y., Feng, J., Chen, H., Lan, J., Zhu, H., Wang, W. and Zhang, J. (2025). WildFake: A Large-Scale and Hierarchical Dataset for AI-Generated Images Detection. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 39, pp. 3500–3508). https://doi.org/10.1609/aaai.v39i4.32363

Huang, G., Liu, Z., van der Maaten, L. and Weinberger, K. Q. (2017). Densely Connected Convolutional Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/1608.06993

Jeong, Y., Kim, D., Min, S., Joe, S., Gwon, Y. and Choi, J. (2022a). BiHPF: Bilateral High-Pass Filters for Robust Deepfake Detection. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. https://arxiv.org/abs/2109.00911

Jeong, Y., Kim, D., Ro, Y. and Choi, J. (2022b). FrePGAN: Robust Deepfake Detection Using Frequency-level Perturbations. In *Proceedings of the AAAI Conference on Artificial Intelligence*. https://arxiv.org/abs/2202.03347

Ju, Y., Jia, S., Ke, L., Xue, H., Nagano, K. and Lyu, S. (2022). Fusing Global and Local Features for Generalized AI-Synthesized Image Detection. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*. https://arxiv.org/abs/2203.13964

Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J. and Aila, T. (2020). Analyzing and Improving the Image Quality of StyleGAN. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/1912.04958

Keita, M., Hamidouche, W., Eutamene, H. B., Taleb-Ahmed, A., Camacho, D. and Hadid, A. (2025). Bi-LORA: A Vision-Language Approach for Synthetic Image Detection. *Expert Systems, 42*(2). https://doi.org/10.1111/exsy.13829

Khan, S. A. and Dang-Nguyen, D.-T. (2024). CLIPping the Deception: Adapting Vision-Language Models for Universal Deepfake Detection. In *Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR)*. https://arxiv.org/abs/2402.12927

Koutlis, C. and Papadopoulos, S. (2024). Leveraging Representations from Intermediate Encoder-blocks for Synthetic Image Detection. In *Proceedings of the European Conference on Computer Vision (ECCV)*. https://arxiv.org/abs/2402.19091

Lin, L., Gupta, N., Zhang, Y., Ren, H., Liu, C.-H., Ding, F., Wang, X., Li, X., Verdoliva, L. and Hu, S. (2024). Detecting Multimedia Generated by Large AI Models: A Survey. *arXiv preprint* arXiv:2402.00045. https://arxiv.org/abs/2402.00045

Liu, Z., Qi, X. and Torr, P. (2020). Global Texture Enhancement for Fake Face Detection in the Wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2002.00133

Liu, H., Tan, Z., Tan, C., Wei, Y., Zhao, Y. and Wang, J. (2024a). Forgery-aware Adaptive Transformer for Generalizable Synthetic Image Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2312.16649

Liu, Z., Wang, H., Kang, Y. and Wang, S. (2024b). Mixture of Low-rank Experts for Transferable AI-Generated Image Detection. *arXiv preprint* arXiv:2404.04883. https://arxiv.org/abs/2404.04883

Luo, Y., Du, J., Yan, K. and Ding, S. (2024). LaRE²: Latent Reconstruction Error Based Method for Diffusion-Generated Image Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2403.17465

Ma, R., Duan, J., Kong, F., Shi, X. and Xu, K. (2023). Exposing the Fake: Effective Diffusion-Generated Images Detection. In *AdvML-Frontiers Workshop at the International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2307.06272

Mandelli, S., Bonettini, N., Bestagini, P. and Tubaro, S. (2022). Detecting GAN-generated Images by Orthogonal Training of Multiple CNNs. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*. https://arxiv.org/abs/2203.02246

Marra, F., Gragnaniello, D., Verdoliva, L. and Poggi, G. (2019). Do GANs leave artificial fingerprints? In *Proceedings of the IEEE Conference on Multimedia Information Processing and Retrieval (MIPR)*. https://arxiv.org/abs/1812.11842

Nataraj, L., Mohammed, T. M., Chandrasekaran, S., Flenner, A., Bappy, J. H., Roy-Chowdhury, A. K. and Manjunath, B. S. (2019). Detecting GAN generated Fake Images using Co-occurrence Matrices. In *IS&T Electronic Imaging: Media Watermarking, Security and Forensics*. https://arxiv.org/abs/1903.06836

Ojha, U., Li, Y. and Lee, Y. J. (2023). Towards Universal Fake Image Detectors that Generalize Across Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2302.10174

Park, J. and Owens, A. (2025). Community Forensics: Using Thousands of Generators to Train Fake Image Detectors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 8245–8257). https://arxiv.org/abs/2411.04125

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G. and Sutskever, I. (2021). Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2103.00020

Rahman, M. A., Paul, B., Sarker, N. H., Hakim, Z. I. A. and Fattah, S. A. (2023). ArtiFact: A Large-Scale Dataset with Artificial and Factual Images for Generalizable and Robust Synthetic Image Detection. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*. https://arxiv.org/abs/2302.11970

Ricker, J., Lukovnikov, D. and Fischer, A. (2024a). AEROBLADE: Training-Free Detection of Latent Diffusion Images Using Autoencoder Reconstruction Error. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2401.17879

Ricker, J., Damm, S., Holz, T. and Fischer, A. (2024b). Towards the Detection of Diffusion Model Deepfakes. In *Proceedings of the International Conference on Computer Vision Theory and Applications (VISAPP)*. https://arxiv.org/abs/2210.14571

Rombach, R., Blattmann, A., Lorenz, D., Esser, P. and Ommer, B. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2112.10752

Schwarz, K., Liao, Y. and Geiger, A. (2021). On the Frequency Bias of Generative Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2111.02447

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-

based Localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. https://arxiv.org/abs/1610.02391

Sha, Z., Li, Z., Yu, N. and Zhang, Y. (2023). DE-FAKE: Detection and Attribution of Fake Images Generated by Text-to-Image Generation Models. In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)* (pp. 3418–3432). https://doi.org/10.1145/3576915.3616588

Simonyan, K. and Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition. In *Proceedings of the International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1409.1556

Tan, M. and Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In *Proceedings of the International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/1905.11946

Tan, C., Zhao, Y., Wei, S., Gu, G. and Wei, Y. (2023). Learning on Gradients: Generalized Artifacts Representation for GAN-Generated Images Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 12105–12114). https://doi.org/10.1109/CVPR52729.2023.01165

Tan, C., Zhao, Y., Wei, S., Gu, G., Liu, P. and Wei, Y. (2024a). Frequency-Aware Deepfake Detection: Improving Generalizability through Frequency Space Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*. https://arxiv.org/abs/2403.07240

Tan, C., Liu, H., Zhao, Y., Wei, S., Gu, G., Liu, P. and Wei, Y. (2024b). Rethinking the Up-Sampling Operations in CNN-based Generative Network for Generalizable Deepfake Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/2312.10461

Tan, C., Tao, R., Liu, H., Gu, G., Wu, B., Zhao, Y. and Wei, Y. (2025). C2P-CLIP: Injecting Category Common Prompt in CLIP to Enhance Generalization in Deepfake Detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*. https://arxiv.org/abs/2408.09647

Tsigos, K., Apostolidis, E., Baxevanakis, S., Papadopoulos, S. and Mezaris, V. (2024). Towards Quantitative Evaluation of Explainable AI Methods for Deepfake Detection. In *Proceedings of the 3rd ACM International Workshop on Multimedia AI against Disinformation (MAD)*. https://arxiv.org/abs/2404.18649

Verdoliva, L. (2020). Media Forensics and DeepFakes: An Overview. *IEEE Journal of Selected Topics in Signal Processing, 14*(5), 910–932. https://doi.org/10.1109/JSTSP.2020.3002101

Wang, S.-Y., Wang, O., Zhang, R., Owens, A. and Efros, A. A. (2020). CNN-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/1912.11035

Wang, Z., Bao, J., Zhou, W., Wang, W., Hu, H., Chen, H. and Li, H. (2023a). DIRE for Diffusion-Generated Image Detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. https://arxiv.org/abs/2303.09295

Wang, X., Guo, H., Hu, S., Chang, M.-C. and Lyu, S. (2023b). GAN-generated Faces Detection: A Survey and New Perspectives. In *Proceedings of the 26th European Conference on Artificial Intelligence (ECAI)*. https://arxiv.org/abs/2202.07145

Wu, H., Zhou, J. and Zhang, S. (2026). Generalizable Synthetic Image Detection via Language-Guided Contrastive Learning. *IEEE Transactions on Artificial Intelligence, 7*(6), 3485–3496. https://doi.org/10.1109/TAI.2025.3641104

Yan, S., Li, O., Cai, J., Hao, Y., Jiang, X., Hu, Y. and Xie, W. (2025a). A Sanity Check for AI-generated Image Detection. In *Proceedings of the International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2406.19435

Yan, Z., Wang, J., Jin, P., Zhang, K.-Y., Liu, C., Chen, S., Yao, T., Ding, S., Wu, B. and Yuan, L. (2025b). Orthogonal Subspace Decomposition for Generalizable AI-Generated Image Detection. In *Proceedings of the International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2411.15633

Yu, N., Davis, L. and Fritz, M. (2019). Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. https://arxiv.org/abs/1811.08180

Zhang, X., Karaman, S. and Chang, S.-F. (2019). Detecting and Simulating Artifacts in GAN Fake Images. In *Proceedings of the IEEE International Workshop on Information Forensics and Security (WIFS)*. https://arxiv.org/abs/1907.06515

Zhu, M., Chen, H., Yan, Q., Huang, X., Lin, G., Li, W., Tu, Z., Hu, H., Hu, J. and Wang, Y. (2023). GenImage: A Million-Scale Benchmark for Detecting AI-Generated Image. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. https://arxiv.org/abs/2306.08571

APPENDIX A: LIST OF ABBREVIATIONS

Table A.1 lists the general abbreviations used in this report. Names of individual datasets, image generators and detection methods (for example GenImage, ProGAN or DIRE) are introduced in the text where they first appear and are not repeated here.

Table A.1. Abbreviations used in the report

Abbreviation	Meaning
AAAI	Association for the Advancement of Artificial Intelligence (conference)
ACC	Accuracy
ACM	Association for Computing Machinery
ADM	Ablated Diffusion Model (a diffusion image generator)
AI	Artificial intelligence
AP	Average precision
AR	Autoregressive
AUC	Area under the curve (here, the ROC curve)
AUROC	Area under the receiver operating characteristic curve
CLIP	Contrastive Language–Image Pre-training
CNN	Convolutional neural network
CSV	Comma-separated values
CVF	Computer Vision Foundation
DCT	Discrete cosine transform
DDIM	Denoising diffusion implicit model
DDPM	Denoising diffusion probabilistic model
DFT	Discrete Fourier transform
DM	Diffusion model
DOCX	Microsoft Word document format (Office Open XML)
DOI	Digital Object Identifier
ECCV	European Conference on Computer Vision
FAR	False alarm rate
FFT	Fast Fourier transform
FN	False negative
FP	False positive
FPR	False positive rate
FR	Functional requirement
GAN	Generative adversarial network
GB	Gigabyte
GPU	Graphics processing unit
Grad-CAM	Gradient-weighted Class Activation Mapping
ICLR	International Conference on Learning Representations

Abbreviation	Meaning
ICML	International Conference on Machine Learning
ID	Identifier
IEEE	Institute of Electrical and Electronics Engineers
JPEG	Joint Photographic Experts Group (a lossy image-compression standard)
KNN	k-nearest neighbours
LDM	Latent diffusion model
LIME	Local Interpretable Model-agnostic Explanations
LPIPS	Learned Perceptual Image Patch Similarity
mAP	Mean average precision
MLP	Multilayer perceptron
NeurIPS	Conference on Neural Information Processing Systems
NFR	Non-functional requirement
NN	Nearest neighbour
NTCC	Non-Teaching Credit Course
Pd@1%FAR	Probability of detection at a 1% false alarm rate
PDF	Portable Document Format
PMLR	Proceedings of Machine Learning Research
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRNU	Photo-response non-uniformity
QF	Quality factor (of JPEG compression)
RAM	Random-access memory
RBF	Radial basis function
RGB	Red, green and blue (colour channels)
RIS	Reverse image search
RISE	Randomized Input Sampling for Explanation
ROC	Receiver operating characteristic
SD	Stable Diffusion
SHAP	SHapley Additive exPlanations
SRM	Spatial rich model (a bank of high-pass noise-residual filters)
SVD	Singular value decomposition
SVG	Scalable Vector Graphics
SVM	Support vector machine
T2I	Text-to-image
TN	True negative
TP	True positive
TPR	True positive rate
ViT	Vision Transformer
VQ	Vector quantisation
VQDM	Vector-quantised diffusion model
WPR	Weekly Progress Report

Abbreviation	Meaning
XAI	Explainable artificial intelligence